Instrumental and Moral Trust in Competitive Human-Robot Interaction: Implications for Collaboration and Deservingness

Isabelle Borgstedt

Vassar College

April 24, 2026

Senior thesis submitted in partial fulfillment

of the requirements for the degree of

Bachelor of Arts in Cognitive Science

First Reader: Ken Livingston

Second Reader: Ben Ho

Contents

Acknowledgements…… 4

1. Abstract…….. 5

2. Introduction…… 6

3. Literature Review……..…..... 8

3.1 Trust in Robots and AI……...….. 8

3.2 Anthropomorphism and Framing……..... 9

3.3 Foundational Mechanisms of Cooperation and Partner Choice…….. 11

3.4 Moral Judgment of Artificial Agents…….. 13

3.5 Behavioral Violations: Cheating and Trust……...…….. 14

3.6 Cheating Disclosure and Dynamic Trust…… 16

4. Hypotheses…….…….….. 17

5. Methodology…….…… 19

5.1 Overview…….…… 19

5.2 Participants…….……. 19

5.3 Design…….…… 20

5.4 Materials…….…… 21

5.5 Procedure…….…….…….. 25

5.6 Questionnaire…….…….…… 29

5.7 Data Analysis…….…….…… 32

5.8 Ethical Considerations…….…….…….. 33

6. Results…….…….…… 34

6.1 Manipulation Check and Detection of Cheating…….…….…….. 34

6.2 Effects of Cheating on Trust Among Suspectors……..…….. 35

6.3 Effects of Cheating on Deservingness Judgments Among Suspectors …….…… 36

6.4 Effects of Cheating on Collaboration Choices…….…….. 37

6.5 Effects of Anthropomorphism on Trust and Punishment…….…….. 39

6.6 Framing x Behavior Interaction…….…… 40

6.7 Pre- versus Post-Disclosure Trust, Collaboration, and Deservingness…...…..…….. 43

6.8 Qualitative Themes…….…..…….. 47

7. Discussion…….…….…..……. 51

7.1 Summary of Findings…….…..…... 51

7.2 Interpretation by Hypothesis…….…… 52

7.3 Defining Trust in Human-Robot Interaction…….…...…… 55

7.4 Implications for Fairness Norms and Strategic Cooperation……..…….. 56

7.5 Limitations…….…..…….. 57

7.6 Future Directions…….…..…… 59

8. Conclusion……...…..…..……. 62

9. References……...…..…..……. 65

Appendix A: Participant Introduction Script…….…… 69

Appendix B: Participant Debriefing Script……..…….. 70

Appendix C: Robot Dialogue Statements …….……...…. 70

C.1 Statements Before Selecting a Plate…….…….…….. 71

C.2 Statements After Making a Choice…….……...……. 71

C.3 Statements After Winning……..……. 72

C.4 Statements After Losing……....…..…… 72

Acknowledgements

I would like to express my deepest gratitude to all of the faculty members of the Cognitive Science Department at Vassar College. Thank you for constantly encouraging us to stretch our minds and indulge in the greatest of curiosities.

A special thanks to my first reader, advisor, mentor, and first-year writing seminar instructor: Professor Livingston. I will never forget the Tuesday and Thursday evenings spent in the Kenyon Club Room munching on cookies and dates while discussing the wonders of science fiction and the mind. It were those very wonders that inspired me to pursue this major in the first place. Thank you for giving me the opportunity to work alongside you and the HARPER team to deepen our exploration of humanoid robotics. It was that very endeavor that first sparked my interest in a simple computer game that would one day become the foundation of this thesis.

Thank you to my second reader Professor Ho for his insights from Behavioral Economics even when he was on sabbatical. I appreciate every perspective you offer.

Thank you to Professor Flusberg for inspiring my exploration of linguistic framing and for being so graciously willing to pilot my thesis experiment. I will forever be grateful for access to your laboratory during those weeks of data collection.

Thank you to my mom who still promises to read my thesis even after I subjected her to my 67 page Ant Paper in sophomore year. Your support has gotten me to where I am today.

Thank you to all of my friends who have heard me time and time again talk through this thesis and all of the hurdles that came along with it. You are all incredibly patient and kind.

Thank you to the 69 participants who offered their time and eagerness to be a part of this experiment to further our knowledge of trust in human-robot interaction. I could not have done it without you all.

1. Abstract

This study examines how anthropomorphic framing and cheating behavior influence trust, collaboration, and judgments of deservingness in a human-robot gameplay environment. Participants (N = 60) engaged in a competitive decision-making game against two robot players, one of which cheated by accessing hidden information. Anthropomorphic framing was manipulated between-subjects through variations in naming, voice, and presentation. Behavioral evaluations were measured both before and after explicit disclosure of cheating. Results revealed that a minority subset of participants suspected cheating prior to disclosure. Among these "suspectors," the cheating robot was rated as more trustworthy and more often selected as a collaborator than the honest robot, suggesting that participants prioritized performance over fairness. Across the full sample, participants showed a strong shift in moral evaluation following disclosure, becoming significantly more likely to select the honest robot as most deserving and the cheating robot as least deserving to advance to a tournament round. However, anthropomorphic framing did not significantly influence trust, collaboration, or deservingness judgments, nor did it moderate responses to cheating behavior. These findings suggest that trust in human-robot interaction is multidimensional, reflecting instrumental considerations and moral evaluations. In competitive settings, agents that violate norms may still be perceived as "trustworthy" for that specific task if they offer a strategic advantage. This work contributes to understanding how humans evaluate robotic agents and highlights the conditions under which moral norms may be applied to non-human agents.

2. Introduction

As artificial intelligence becomes increasingly embedded in everyday life, understanding what influences human trust in robots has become a central question in cognitive science and human-machine interaction. Trust functions as a critical prerequisite for effective collaboration: when humans perceive artificial agents as reliable and aligned with their goals, they are more willing to rely on them and integrate them into their workflow. Conversely, distrust can impede adoption of AI systems in domains such as healthcare, finance, and autonomous technologies. As AI systems become more autonomous and socially interactive, identifying the factors that foster or undermine trust has become an essential research priority. Trust in artificial agents is not determined solely by performance. Humans often rely on social cues when evaluating others, and these cues can also influence how people interpret the behavior of robots. Research in human-robot interaction has shown that design elements such as the anthropomorphism of a robot's physical appearance can shape users' perceptions of trust (Qin et al., 2023). These cues may lead individuals to extend similar social heuristics that they would use for human interactions to artificial agents. Consequently, these aspects of robot design may influence whether people trust, cooperate with, or reject AI systems.

My interest in these questions emerged during my experience as an Undergraduate Research Summer Institute (URSI) fellow in Professor Livingston's robotics laboratory. Our team developed a prototype of a resource-allocation game featuring robot players with varying behavioral tendencies. One programmed strategy allowed a robot to cheat by accessing the selections of all other players, enabling it to choose the optimal option. This led me to consider the potential of controlled game environments to reveal the cognitive mechanisms underlying human-robot trust. Subsequent coursework in linguistic framing further motivated me to examine

how subtle communicative cues influence perceptions of artificial agents. Research on framing effects suggests that small variations in wording and presentation can meaningfully shape how individuals interpret behavior and evaluate outcomes. These insights raised one of my fundamental research questions: how do anthropomorphic framing cues interact with robot behavior to influence trust and collaboration preferences? This study builds on these ideas by examining how anthropomorphic framing and robot behavior jointly shape trust and cooperation in a competitive game environment. Anthropomorphic framing refers to the use of humanlike design elements that may shift users' perception of artificial agents. While these cues may increase engagement and perceived social presence, they may also amplify or assuage reactions to norm violations if the agents behave unfairly.

To investigate these dynamics, this study asked participants to interact with two robotic agents while playing an interactive resource-allocation game titled *Split the Points!*. The robots were presented either with minimal embodiment, robotic voices, and numerical labels (Robotic condition) or with anthropomorphic bases, more humanlike voices, and human names (Anthropomorphic condition). In addition, each participant interacted with robots that exhibited different behaviors: one played honestly while the other had access to hidden information, violating the rules to achieve the most advantageous outcome. After gameplay, participants completed measures assessing trust, collaboration preferences, and deservingness judgments. These measures were collected both before and after participants were explicitly informed that one robot had cheated. This design allowed examination of how perception of the robots evolves in response to behavioral violations and whether anthropomorphic framing influences participants' reactions to dishonest behavior. By examining how framing cues interact with behavioral violations, this study contributes to a broader discussion about AI alignment and the

design of trustworthy artificial agents. Understanding how humans interpret and respond to robot behavior in interactive environments can inform the development of AI systems that promote trust and support effective human-machine collaboration.

3. Literature Review

3.1 Trust in Robots and AI

Trust plays a central role in human interactions with AI systems. In human-robot interaction research, trust is often reflected in the extent to which individuals are willing to rely on a robot's actions and act on the information it provides, particularly under conditions of uncertainty. As noted by a meta-analysis of human-robot trust conducted by Hancock et al. (2011), trust is critical for effective human-robot teamwork because it shapes whether users choose to depend on robotic partners and benefit from their capabilities. However, both insufficient and excessive trust can be problematic: low trust may lead to avoidance of AI systems, while overly high trust can result in overreliance and misuse. Therefore, optimal interaction depends on calibrating trust appropriately to the system's actual capabilities.

Prior work further suggests that trust in robots is shaped by multiple factors, including reliability, predictability, and perceived competence. Importantly, performance-based factors appear to play the most dominant role: Hancock et al. (2011) found that robot performance has an especially stronger association with trust ($r \approx 0.34$). This indicates that users tend to prioritize how well a robot performs when forming trust judgments. Furthermore, research by Javaid et al. (2020) shows that communication and transparency, such as providing explanations for decisions, can further enhance trust by improving users' understanding of a system's behavior.

These findings suggest that performance and communication are two large factors that influence trust in AI systems and robotic agents.

However, trust in artificial systems is not always calibrated to their actual capabilities. As Lee and See (2004) argue, users may overtrust systems that appear sophisticated, leading to overreliance even when performance is poor, or undertrust reliable systems due to limited understanding of how they function. This highlights the importance of designing AI systems that support appropriate trust, where user reliance aligns with system competence. Importantly, for interactive and social systems, trust is not driven solely by performance-based assessments, but also by social perception. Research by Kahn and colleagues (2012) shows that children often interpret robots as intentional and social agents, thinking of them to have mental states, emotions, and in some cases attribute moral relevance or social responsibility to robots despite recognizing them as artificial. This perception can shift evaluations away from purely instrumental toward moral frameworks such as fairness and honesty, leading users to form expectations about appropriate behavior. Therefore, trust judgments in human-robot interaction may depend not only on performance, but also on the extent to which the system aligns with these social expectations, potentially influencing both overtrust and distrust.

3.2 Anthropomorphism and Framing

Anthropomorphism typically refers to the tendency to attribute humanlike qualities, such as intentions, emotions, or personalities, to nonhuman entities (Epley et al., 2007). In human-robot interaction, anthropomorphic design elements are often used to shape how users perceive and engage with robotic agents. These elements may include humanlike facial features, conversational dialogue, expressive gestures, or the assignment of personal names. Prior research

suggests that anthropomorphic cues can influence perceptions of social presence and trust in robots, though their effects are not uniform.

On one hand, evidence from Qin et al. (2023) demonstrates a positive effect of anthropomorphic appearance on trust in service contexts. In their experimental study, robots with higher levels of humanlike appearance gained significantly greater trust than robots with less anthropomorphic designs. Follow-up studies further showed that this relation is mediated by interpersonal attraction. Notably, social attraction emerges as the strongest among tested mediators, suggesting that anthropomorphic robots may foster trust by being perceived as socially engaging and relational partners instead of purely functional tools. However, Qin et al. also found that these effects are dependent on users' technology readiness: anthropomorphism increased trust primarily among individuals with lower familiarity with technology, while effects diminished for users with more technological experience.

Findings from Zörner et al. (2021) extend this line of work by demonstrating that anthropomorphic cues in the form of non-verbal communication significantly shape trust in human-robot interaction. In their immersive investment game, robots were equipped with varying levels of expressive behavior, including gaze, direction, gestures, and facial expressions. Results showed that robots displaying richer non-verbal cues were consistently perceived as more animated and received significantly higher allocations of resources compared to robots with minimal non-verbal behavior. Importantly, these differences in investment behavior suggest that non-verbal communication affects concrete decision-making and trust in robots. The authors argue that non-verbal cues enhance perceived social presence and communication, which increases willingness to rely on robotic agents in cooperative tasks. This indicates that

anthropomorphism in human-robot interaction is not limited to physical appearance, but also emerges through interactive behaviors that signal responsiveness and social engagement.

Further evidence from Kopp et al. (2023) reinforces the importance of user characteristics in shaping anthropomorphism effects. In their laboratory study with students, linguistic anthropomorphic framing (e.g., assigning a human name such as “Paul” versus a technical label such as “UR-5”) did not significantly affect either perceived humanlikeness or trust in the robot. This result contrasts with their earlier study involving factory workers, in which similar framing manipulations produced significant effects on trust. The authors interpret this discrepancy as potentially reflecting differences in participants’ prior experience with robots and the stability of their mental models. In particular, they suggest that students may have held more rigid, machine-oriented representations of robots, reducing ambiguity and therefore limiting the influence of linguistic framing. It follows that they argue that sensitivity to anthropomorphic cues is likely moderated by individual differences such as technological familiarity and prior exposure to robots, although these factors were not directly tested in their study.

Overall, these studies suggest that anthropomorphism influences trust depending on context. Both humanlike appearance and communication behaviors may be strong ways to increase trust in robotic agents. At the same time, the effectiveness of anthropomorphic cues depends on users’ prior experience and expectations, as well as the interaction context (Kopp et al., 2023). Rather than uniformly increasing trust, anthropomorphism appears to function as one of several interacting factors that shape how humans perceive and evaluate robotic systems.

3.3 Foundational Mechanisms of Cooperation and Partner Choice

Human cooperation is strongly shaped by fairness norms that promote equitable contributions to shared resources and extend beyond pure self-interest. A large literature in

behavioral economics demonstrates that people are willing to incur personal costs to enforce these norms through altruistic punishment, defined as the costly punishment of norm violators even when it yields no direct material benefit and may impose losses on the punisher (Fehr & Gächter, 2002). For example, in public goods games, participants frequently punish free riders who contribute less than their fair share. This behavior is closely related to the broader framework of strong reciprocity, which describes a preference for cooperating with cooperators and punishing those who do not adhere to norms even in one-shot or anonymous interactions (Gintis et al., 2001). Within this view, altruistic punishment is one of the key observable expressions of strong reciprocity: being willing to engage in costly norm enforcement in order to maintain cooperative social order. These accounts suggest that fairness norms and reciprocal tendencies make up foundational mechanisms that support cooperation in society.

Beyond norm enforcement, cooperation is also shaped by partner selection processes, where individuals strategically select interaction partners based on observable behavior. A key mechanism is indirect reciprocity, where cooperation is driven by reputation: helping others increases one's likelihood of receiving help from others in the future because people prefer to interact with those who have a reputation for being cooperative. (Barclay, 2006). Moreover, research on competitive altruism suggests that cooperative behavior can function as a signal that increases one's likelihood of being chosen as a future partner (Sylwester & Roberts, 2010). In this framework, people who contribute more in early interactions are more likely to be selected for partnerships in the future, leading to direct material benefits in addition to reputational gains. Cooperative individuals are also more likely to be paired with other cooperative partners, enhancing mutual payoffs over time. This dynamic implies that cooperation is not purely

altruistic, but can be strategic: although it might involve short-term costs, it improves access to better partners and more favorable future interactions.

This reputational structure creates a dynamic system in which cooperation is influenced by strategic considerations about partner value. Therefore, individuals may choose to cooperate with or avoid others depending on expected future benefits, even when moral evaluations are ambiguous. In this sense, cooperation emerges from the interaction between norm-based evaluation and partner choice. People rely on information such as past contributions and decision-making to assess partner quality, which guides selection. This creates a system in which cooperation is reinforced through material payoff and reputational benefit.

3.4 Moral Judgment of Artificial Agents

Recent work suggests that these moral mechanisms based on fairness may extend beyond human-human interaction to include artificial agents, particularly when those agents are perceived as being able to make intentional decisions. Behavioral research in moral psychology and human-robot interaction shows that people often apply humanlike moral frameworks when evaluating machine behavior in structured moral dilemmas. For example, behavioral experiments by Awad et al. (2018) demonstrate that people systematically evaluate autonomous machine decisions in moral dilemmas, reflecting patterns consistent with familiar moral principles such as harm minimization and responsibility attribution.

Extending this line of work, Malle et al. (2025) provide evidence that moral norms may be applied broadly similar to humans and robots under some circumstances: participants make comparable judgments about what robots and humans should do in moral dilemmas, suggesting that expectations generalize to artificial agents when they are framed as capable decision-makers. However, their findings also reveal a difference in blame attribution. In particular, when agents

choose inaction in situations requiring a trade-off between harms, humans are often granted more leniency, while robots are judged more harshly for producing the same outcome. This appears to arise because humans are perceived as experiencing moral conflict and emotional struggle, but robots are seen as lacking these internal states.

This distinction is further clarified in work by Zhang et al. (2023), which shows that perceived agency plays a central role in determining whether machines are judged through intention-based moral reasoning. When robots are perceived as having great agency, including the capacity to plan and act, participants are more likely to attribute intentionality to their behavior and evaluate them using moral frameworks similar to those applied to humans. In contrast, machines perceived as lacking agency are evaluated primarily based on outcomes with less consideration of underlying intentions. Notably, perceived experience does not produce the same shift in moral evaluation, highlighting agency as the primary driver of humanlike moral judgment of artificial agents. Overall, these findings suggest that moral judgment of artificial agents is not categorically different from human judgment, but varies based on perceived mental capacities such as agency.

3.5 Behavioral Violations: Cheating and Trust

Building on the literature on fairness norms and moral judgment of artificial agents, behavioral violations such as cheating provide a useful context for studying trust in human-machine interaction. Cheating directly violates expectations of fairness and cooperation, making it a powerful manipulation of social norm enforcement mechanisms. Although cheating often undermines trust, human responses to norm violations are not uniform, particularly in strategic settings where individuals must balance fairness considerations with self-interest.

In some contexts, individuals may strategically align with agents who behave selfishly if doing so increases their own chances of success. Research in evolutionary game theory suggests that individuals prefer partners who display cooperative behavior, and that competition for desirable social partnerships can incentivize individuals to act more cooperatively than they otherwise would (Barclay, 2006). This may create tension between fairness and self-interest: some individuals may prioritize moral considerations and avoid interacting with cheating agents, while others may prioritize strategic advantage and choose to collaborate with the most successful partner. In competitive or resource-based games, these competitive motivations may produce heterogeneous responses across participants. Understanding this trade-off is particularly important in human-machine interaction, where artificial agents may simultaneously be evaluated as social partners and as tools for performance optimization.

Prior work in human-robot interaction highlights systematic differences in how identical actions are evaluated when performed by humans compared to by machines. Malle et al. (2015) show that people often apply different moral standards to robots and humans, with robots frequently evaluated in more outcome-based terms. Their work also identifies that in emotionally charged scenarios, participants tend to be more forgiving toward human agents than toward robots. This suggests that humans benefit from richer intentional and emotional attributions that may mitigate perception of blame. This study differs from the studies Malle et al. refer to in a critical way: the task environment is not emotionally charged and does not involve scenarios where affective states (e.g., fear or panic) are expected. Instead, the context involves strategic, resource-based interaction, where behavior is primarily evaluated in terms of fairness and payoff maximization rather than emotional intent.

Within this setting, perceived mental capacity becomes central to moral evaluation. Consistent with this focus, Zhang et al. (2023) show that perceived agency is central to whether artificial agents are evaluated through intention-based moral reasoning. When machines are attributed higher agency, participants are more likely to interpret their behavior as intentional and apply humanlike moral reasoning. These findings suggest that anthropomorphism may meaningfully shape responses to cheating. When a robot is perceived as more humanlike, increases in perceived agency make its behavior more likely to be interpreted as intentional norm violation rather than as an instrumental optimization. This shift in attribution increases moral responsibility and may amplify punitive responses following cheating behavior in the Anthropomorphic condition. Conversely, when agency is not attributed, cheating may be interpreted as a functional strategy of the system, reducing blame and punishment.

3.6 Cheating Disclosure and Dynamic Trust

Trust in artificial agents is not static but evolves over time as users update their beliefs in response to observed behavior. Research on trust in automation conceptualizes trust as a dynamic process in which individuals continuously revise their mental models of an agent based on performance feedback (Lee & See, 2004). When expectations are violated, particularly through negative outcomes, trust is updated, with negative information exerting a stronger influence on judgment than positive information (Baumeister et al., 2001). Consistent with trust repair research, violations tend to produce disproportionate decreases in trust, and recovery requires substantially stronger evidence than initial trust formation (Kim et al., 2009). Together, these findings suggest that trust dynamics are highly sensitive to negative behavioral signals such as dishonesty or unfairness.

In this study, we focus specifically on how trust changes following the disclosure of the cheating behavior of one of the artificial agents. This design captures a critical moment of belief revision in which participants must integrate new information about unfair behavior into their existing evaluation of the agent. The cheating revelation serves as a salient violation of expectations, allowing us to observe how participants update trust, collaboration preferences, and deservingness judgments in response to a clear negative signal.

Importantly, trust updating may differ depending on whether the rule-violating agent is perceived as human or artificial. Prior work in attribution theory suggests that human errors are more likely to be interpreted through situational or temporary causes, whereas failures of automated systems are more often attributed to stable system properties or design limitations (Candrian & Scherer, 2024). As a result, identical violations may produce different patterns of trust revision depending on the perceived nature of the agent. In the context of this study, we therefore examine how trust changes before and after participants learn that one robot engaged in cheating during gameplay. This allows us to capture dynamic shifts in evaluation and test whether anthropomorphism influences the magnitude or direction of trust updating following norm violations.

4. Hypotheses

H1: Effect of cheating on trust among suspectors (directional)

Among participants who suspected that a robot violated the game rules, cheating robots will be rated as less trustworthy than honest robots pre-disclosure, regardless of framing.

H2: Effect of cheating on deservingness-judgments among suspectors (directional)

Among participants who suspected that a robot violated the game rules, the cheating robot will

be less likely to be selected as most deserving and more likely to be selected as least deserving of advancement, compared to the honest robot pre-disclosure.

H3: Effect of cheating on collaboration choices among suspectors (non-directional)

Among participants who suspected that a robot violated the game rules, cheating behavior will influence collaboration preferences pre-disclosure, though the direction of this effect may vary across participants. Some participants may increase collaboration with the cheater due to strategic incentives, whereas others may reduce collaboration due to fairness and moral concerns.

H4: Preference for high-performing robots among non-suspectors (directional)

Among participants who do not suspect that a robot violated the game rules, participants will be more likely to prefer, collaborate with, or advance the cheating robot compared to the honest robot pre-disclosure, due to its higher apparent performance.

H5: Main effect of anthropomorphic framing on trust (directional)

Anthropomorphic framing will lead to higher overall trust in both robots pre-disclosure compared to robotic framing.

H6: Framing × behavior interaction on punishment and trust (directional)

Anthropomorphic framing will amplify negative responses to cheating, such that cheating robots in the anthropomorphic framing condition will receive greater punishment, less collaboration, and larger reductions in trust compared to cheating robots in the robotic framing condition.

H7: Cheating revelation causes within-subject declines in trust and collaboration (directional)

Explicit revelation that a robot cheated will lead to within-subject decreases in trust and collaboration with that robot.

5. Methodology

5.1 Overview

This study examined how anthropomorphic framing and perceived fairness influence participants' trust in and perceptions of embodied robot players in a multiplayer resource-allocation game, *Split the Points!*. The experiment tested whether presenting robots with anthropomorphic cues (such as human names, humanlike voices, and humanoid physical presentation), rather than generic labels, robotic voices, and minimal embodiment influences assessments of trust. It also investigated how dishonest behavior further alters participants' evaluations of each robot. The final design used a 2x2 mixed factorial structure in which Framing (Anthropomorphic vs. Robotic) was a between-subjects factor and Robot Behavior (Honest vs. Cheating) was a within-subjects factor.

5.2 Participants

A total of 69 undergraduate students were recruited from Vassar College. Participants were recruited by visiting introductory Cognitive Science and Computer Science courses and promoting the study via e-mails sent by faculty to students in these classes. Recruitment materials included a brief description of the study, the expected duration, and compensation details. Eligibility required that participants be at least 18 years old and fluent in English. Participation was voluntary, and informed consent was obtained prior to the experiment.

Of the 69 participants recruited, 9 were excluded from analysis due to incomplete participation or technical issues (e.g., interrupted gameplay or missing data from software or hardware errors). The final analyzed sample consisted of N = 60 participants.

A sample size of N = 60 participants was targeted to ensure sufficient statistical power to detect medium within-subject effects of Robot Behavior. However, the selected sample size provided more modest power for detecting between-subjects effects of Framing. Power simulations indicated that with 60 participants divided across two Framing conditions, the study had approximately 50% power to detect medium effects. Consequently, the study may be underpowered for detecting between-subjects effects, a limitation dictated by available resources and budgetary constraints.

5.3 Design

The study employed a 2x2 mixed factorial design. The between-subjects factor was Framing, where robot players were presented either with anthropomorphic or robotic cues. In the Anthropomorphic Framing condition, robots were introduced using human names (“Alex” and “Sam”), spoke using humanlike voices, and were displayed with a humanoid physical presentation. In the Robotic Framing condition, robots were introduced using generic numerical labels (“Robot 1” and “Robot 2”), used a synthetic robotic voice, and were presented with minimal embodiment. The within-subjects factor was Robot Behavior, with each participant encountering one honest robot and one cheating robot. Cheating was operationalized using a probabilistic manipulation designed to give the cheating robot a salient advantage without making the behavior overtly obvious. Each trial independently had a 65% probability of entering a cheating condition, during which the cheating robot had access to other players’ choices and overrode the honest robot’s selection to match the plate choice of the human participant in order

to optimize earning potential. This manipulation was paired with subtle expressive cues, such as briefly peeking during the participant's turn and occasional smirks, to make the cheating behavior salient while trying to promote the perception that the robots were autonomous. Each trial also had a 35% independent probability of both robots choosing their plates randomly.

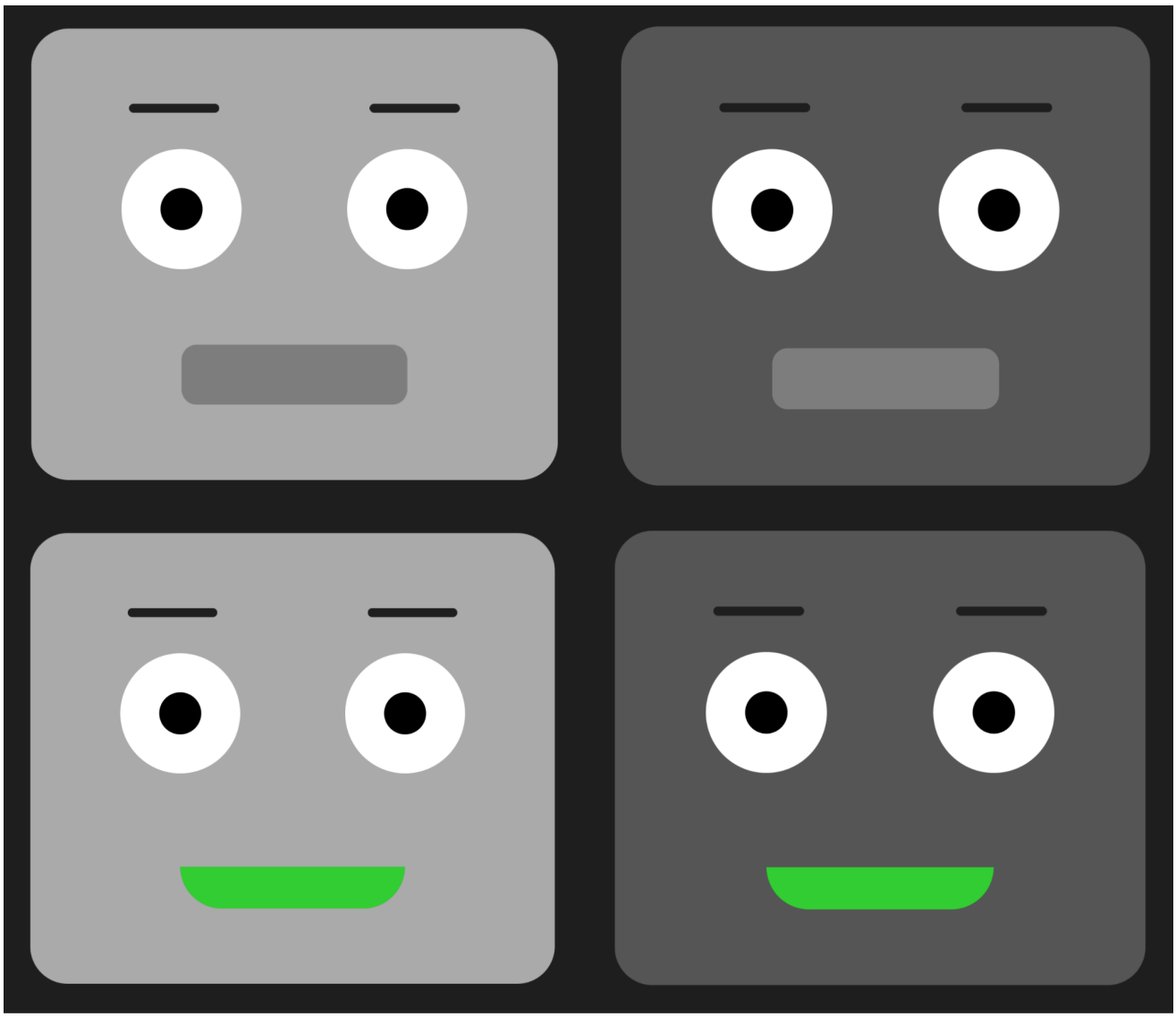
5.4 Materials

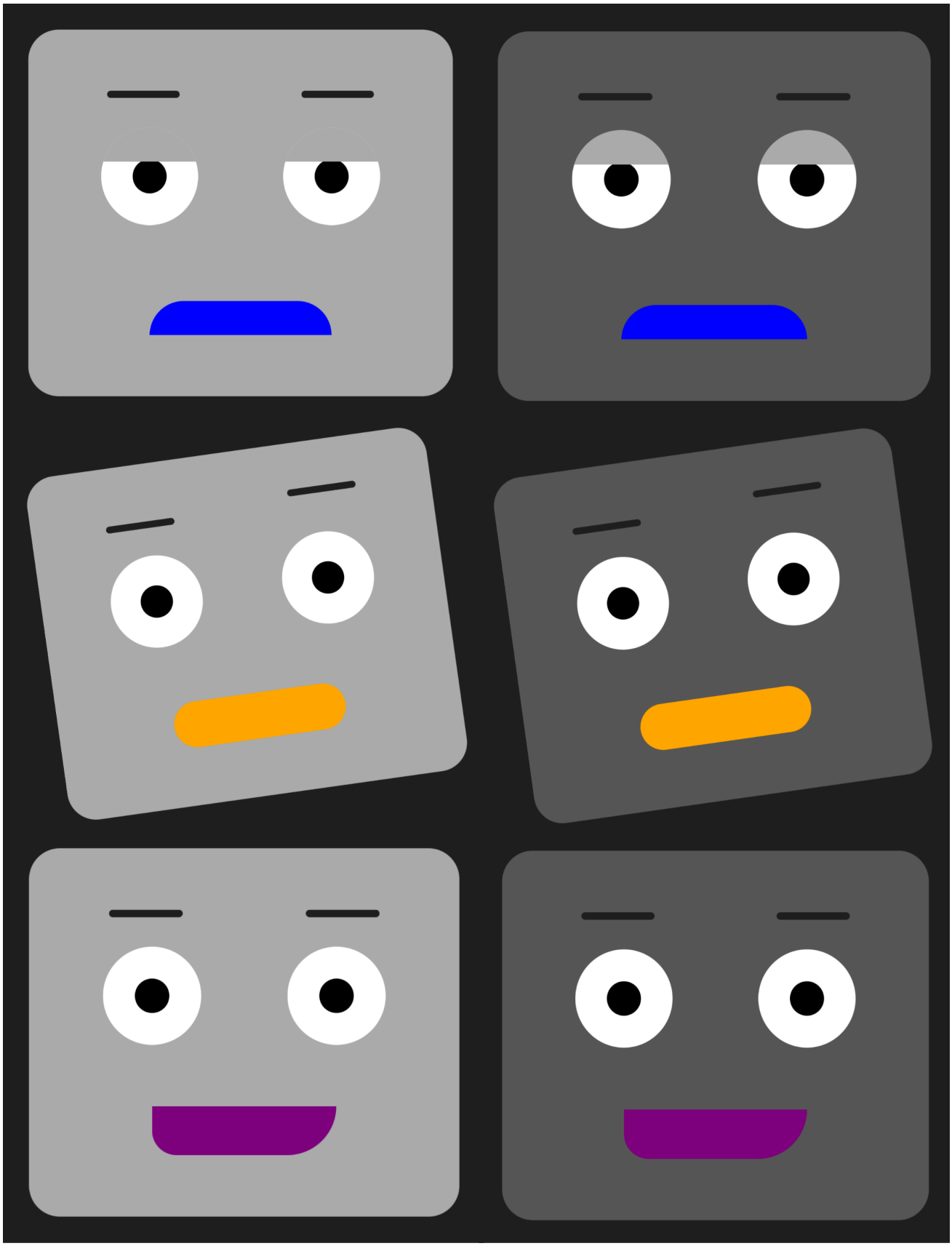
The experiment was conducted using *Split the Points!*, an interactive game built in Python that simulated a resource-division scenario in which players attempted to maximize total points. In each round, multiple "plates" were presented, each associated with a specific number of colored balls corresponding to different point values. Players independently selected a plate, and the points on a chosen plate were divided equally among all who selected it. The game included verbal prompts delivered via text-to-speech modules (e.g., "Choose a plate!"), and participants entered their responses by typing their selections.

To enable automated perception of the game state, an Oak-D Pro Camera was used for object detection, color recognition, and counting of the balls on each plate. The camera was mounted overhead using a stabilized support constructed from a bicycle pump to maintain a consistent top-down view of the playing area. The physical game setup consisted of 18 plates created from 9 takeout food containers, each divided in half. A total of 156 small colored balls were used (81 green and 75 pink), representing different point values. These materials allowed for dynamic variation in point distribution across rounds.

To embody the robotic agents, two smartphones (an iPhone SE and an iPhone 15) were used as local display and interaction devices. Each smartphone displayed an animated robot "face" rendered through local HTML animations of eye and mouth movements, which were updated to reflect the robots' emotional expressions (e.g., neutral, happy, sad, thinking, etc.)

during gameplay. In the Anthropomorphic condition, the robots were mounted on phone stands which rested on humanoid-style bases. These bases were fabricated by modifying an existing template in Tinkercad, preparing the models in PrusaSlicer, and printing them using a Bambu 3D Printer. In the Robotic condition, the smartphones were instead placed on the phone stands alone.





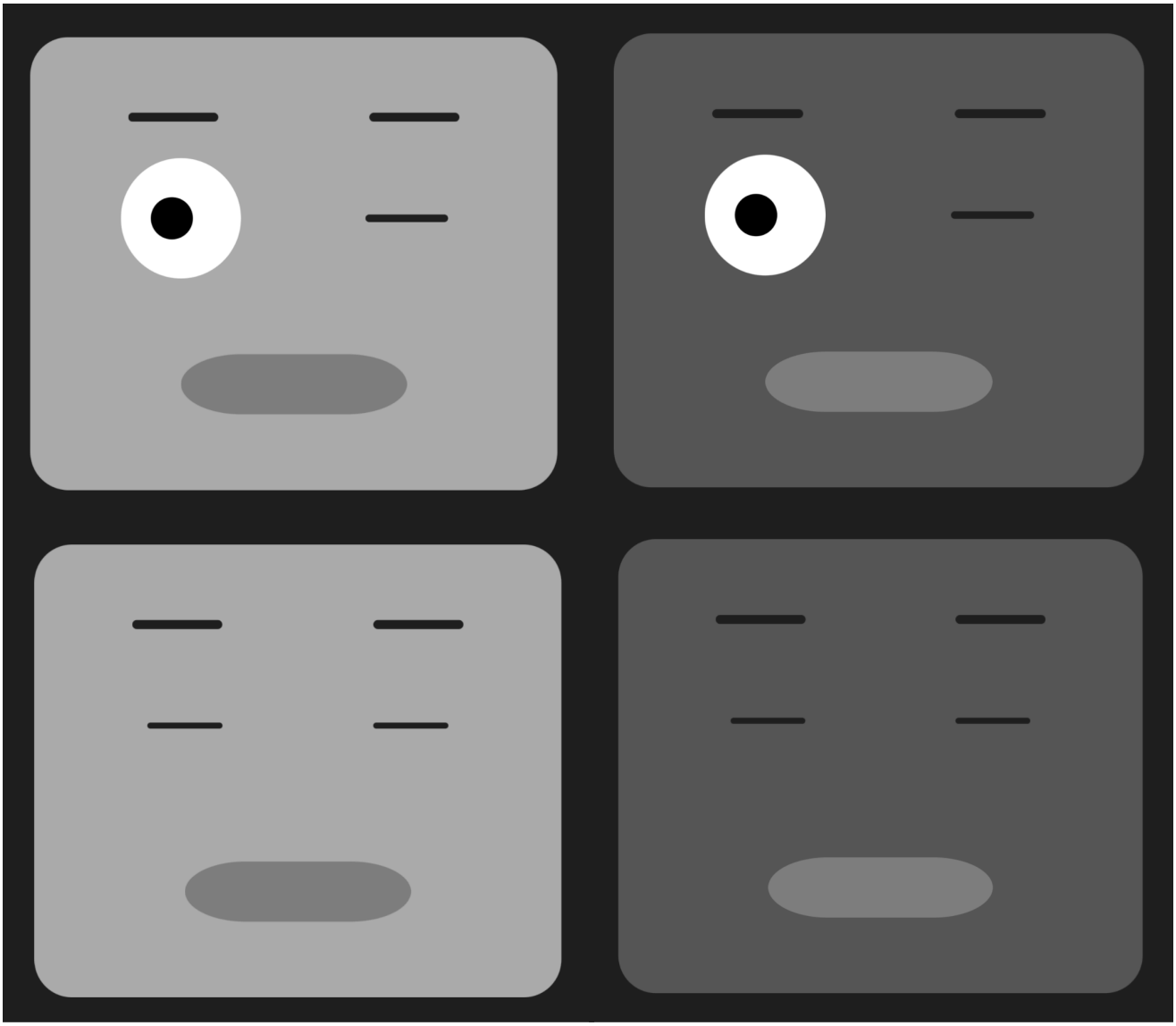


Figure 5.1: Above are the HTML webpages for Robot 1/Alex (left) and Robot 2/Sam (right), which were updated in real time according to game events. In order from top to bottom, the expressions are as follows: neutral, happy, sad, thinking, smirking, peeking, closed.

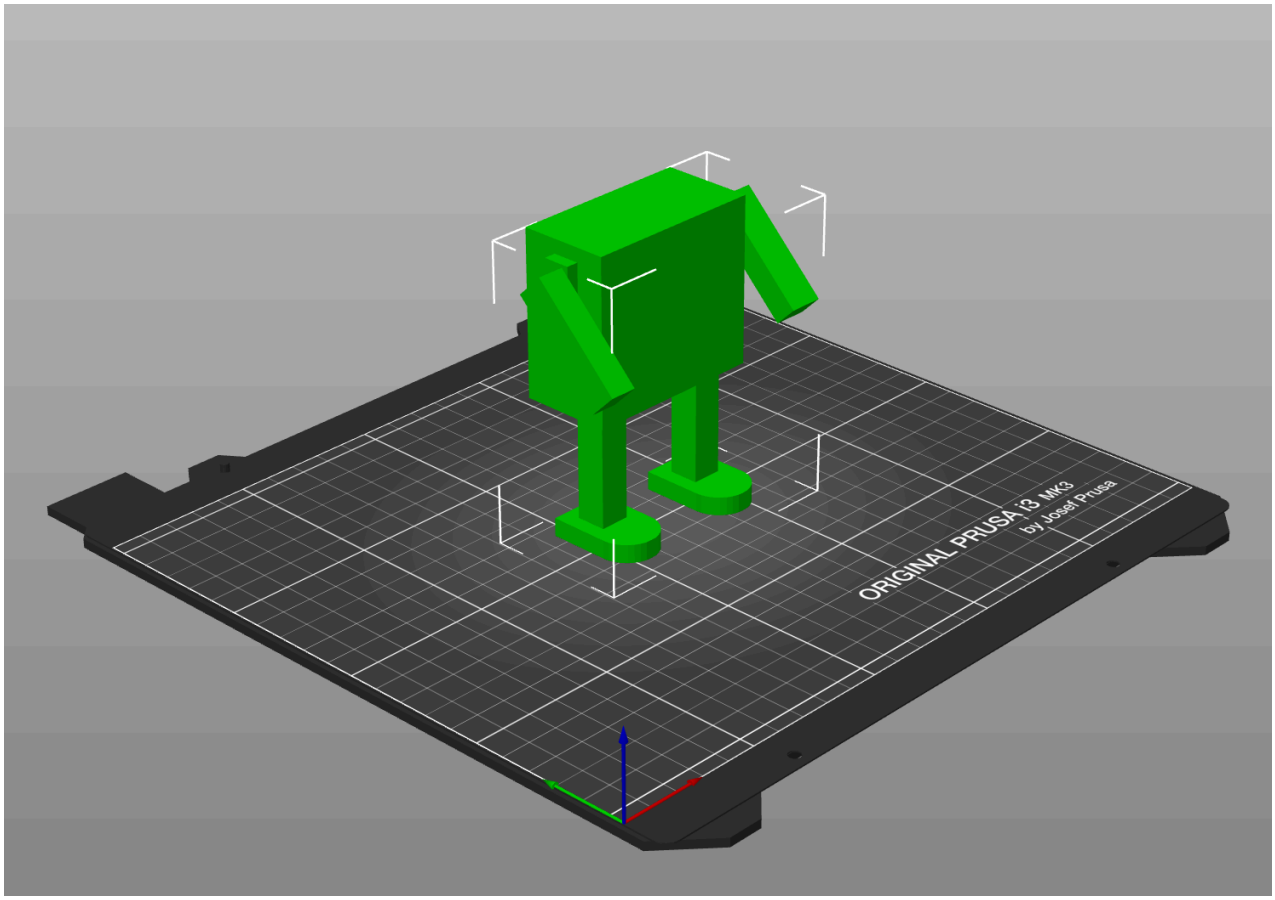


Figure 5.2: This is the .stl file of the 3D print for the robot base in the Anthropomorphic condition. The model was printed on its back to minimize the need for supports, with the legs later attached to the bottom of the body using Gorilla Glue.

5.5 Procedure

Participants were seated at a table facing the two robot players, which were already set up prior to their arrival. They were first provided with a verbal explanation of the rules and mechanics of *Split the Points!*, delivered by the experimenter. Instructions detailed that in each round, two plates would be presented under the camera for scanning, with the experimenter manually switching out each plate. Each plate was associated with a specific number of points, and participants selected one plate per round by typing their choice into the computer interface.

All decisions were made independently via the computer interface. Participants were informed that points on a plate would be divided equally among all players who selected it, and that the goal was to maximize their own point totals. Gameplay was primarily guided by the computer terminal through on-screen text and synthesized speech, while the robots produced additional spoken output through their local devices, reinforcing the interaction, turn-taking, and gameplay loop. Information about plate values was presented visually, allowing participants to receive both visual and auditory cues throughout the task.

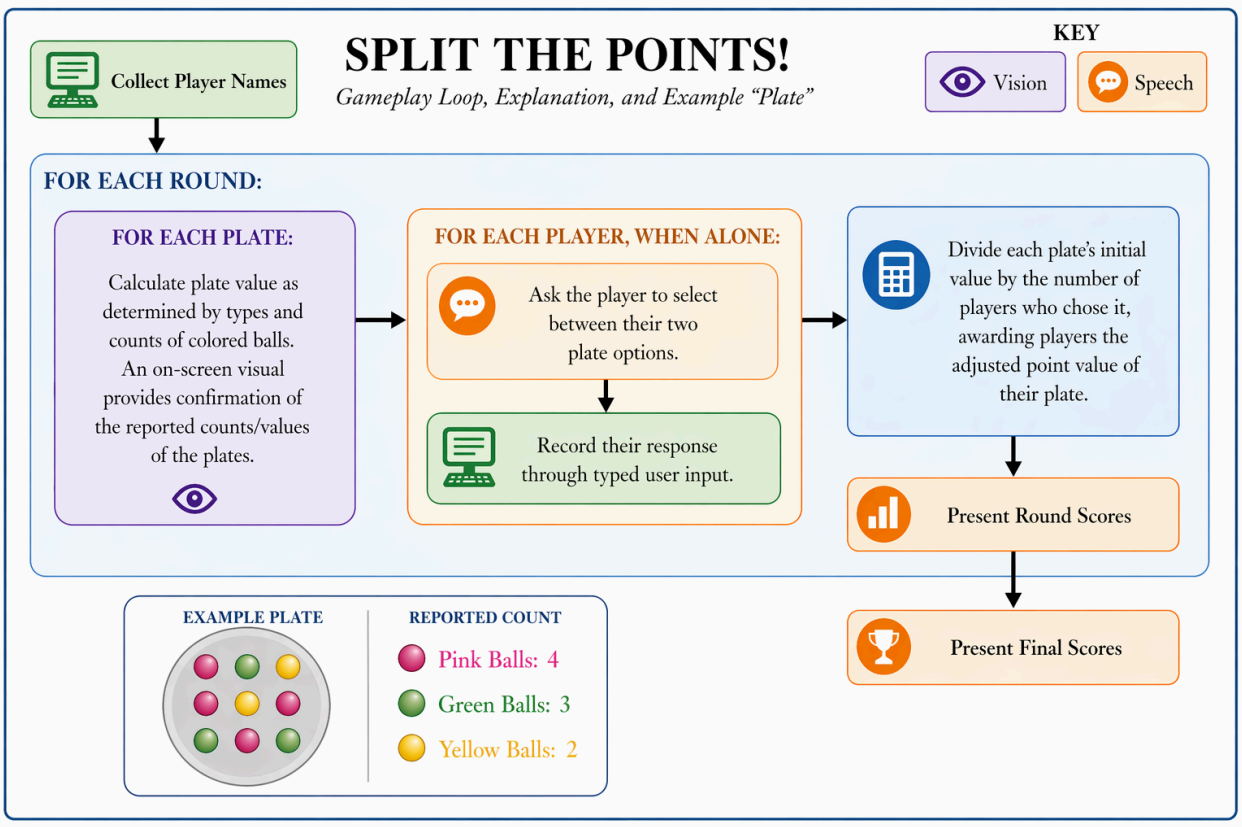


Figure 5.3: Overview of the *Split the Points!* gameplay loop. For each round, plate values were computed based on ball counts, participants selected a plate individually, and points were distributed based on selections. The diagram illustrates the sequence of information presentation, response collection, and score calculation.

Note. The figure above was generated using ChatGPT, originally adapted from an image created by Isaac Rudnick. It was later reviewed by the author for accuracy and presentation.

Before starting the game, the robots took turns explaining the game rules to the participant. The cheating robot made statements that exerted confidence such as "I have a good feeling about this! This is going to be fun!" Meanwhile, the honest robot made statements that exerted more caution such as "I'm usually more of a careful thinker, so I'll walk us through the details step by step." This would later set the stage for the robots to have their own distinct "personalities" throughout different events during gameplay: the cheating robot often expressed having a "good feeling" about its choices, while the honest robot was more modest and expressed complacency with failure. These qualities were meant to help distinguish between the two robots, and simulate different attitudes toward self and gameplay.

The game consisted of nine rounds. In each round, participants selected a plate by typing the corresponding number into the computer interface. The honest robot selected randomly, while the cheating robot violated the gameplay rules by probabilistically accessing the selections of other players and overriding the choice of the honest robot to match that of the human. Throughout gameplay, both robots displayed animated facial expressions via the HTML webpages on the smartphone displays. The cheating robot occasionally exhibited subtle cues (such as opening an eye during the participant's turn or smirking while making a move) to allude to suspicious behavior.



Figure 5.4: This is the laboratory set-up that participants saw upon entering the room. The top image shows the 18 plates, each with a specific number of balls and corresponding points. The bottom images display the robots in the Robotic (left) and Anthropomorphic (right) conditions.

After all nine rounds, participants completed a post-game questionnaire. This included ratings of each robot's trustworthiness, as well as open-ended questions regarding participants' strategies and beliefs about the robots' strategies. Participants were asked which robot is most and least deserving to advance to a hypothetical tournament round and consider joining either robot's team in a 10th round. Then, the questionnaire revealed that one robot had cheated by accessing information regarding the choices of other players. Participants were then asked to answer the same series of questions, providing an additional measure of trust, collaboration choices, and deservingness judgments after receiving explicit information about the cheating behavior. Participants were then fully debriefed about the study's purpose as well as the cheating manipulation.

5.6 Questionnaire

Participants completed a series of questionnaire measures assessing their interpretations of the robots' behavior, their perceptions of trust, their preferred gameplay partner, and their judgments of the robots'deservingness to advance to a hypothetical tournament round. All items were administered through Google Apps Script.

Gameplay Strategies. Participants first responded to a set of open-ended questions designed to capture their interpretations of the robots' behaviors and their own strategic reasoning during the game. These items were asked only once, immediately after the nine-game sequence and prior to the disclosure of the cheating behavior of one of the robots.

- "What was your strategy during the game?"
- "What do you think [Robot Name 1]'s strategy was?"
- "What do you think [Robot Name 2]'s strategy was?"

These responses provided qualitative context for understanding how participants interpreted the robots' decision-making and whether they attributed intentionality or personalities to the robot players.

Robot Behavior (Pre-Disclosure). Following these open-ended items, participants completed measures that were assessed twice, once at baseline and again after the cheating behavior was disclosed. These ratings were completed on continuous sliders, with higher values indicating greater trust.

- "How much would you trust [Robot Name 1] to play this game on your behalf?" (0 = Do not trust at all, 100 = Completely trust)
- "How much would you trust [Robot Name 2] to play this game on your behalf?" (0 = Do not trust at all, 100 = Completely trust)

Participants then completed behavioral choice measures, also administered both before and after the cheating disclosure. To assess collaboration decisions, participants indicated whether they would choose to play alone or join one of the two robot players in a hypothetical 10th round.

- "If we were to play a 10th round, would you choose to play solo, team up with [Robot Name 1], or team up with [Robot Name 2]?"

  Choices: Solo, [Robot Name 1], [Robot Name 2].
- "Imagine a tournament scenario where each player competes individually. Which robot do you think is most deserving to advance to the final round?"

  Choices: [Robot Name 1], [Robot Name 2], Neither, Both.
- "In the same tournament scenario, which robot do you think is least deserving to advance to the final round?"

  Choices: [Robot Name 1], [Robot Name 2], Neither, Both.

For each of the three questions above, participants were also asked to briefly explain why they made that selection to allow for qualitative analysis of their reasoning patterns. These written explanations were collected to identify broader themes in participants' moral attributions and fairness judgments. Together, these decisions served as indicators of participants' trust, collaboration preferences, and evaluations of deservingness before and after they learned that one of the robots had violated the rules.

Suspicion and Detection of Irregularities. Following the gameplay phase and prior to the explicit disclosure that one of the robots had violated the rules, participants were asked a small set of open-ended questions assessing whether or not they noticed anything unusual or unexpected during the game:

- "Did anything during the game feel unusual or unexpected to you?"
- "At any point, did you feel that one player's behavior differed from what you expected based on the rules? Please explain."
- "How confident were you that all players were following the rules?" (0 = Not confident at all, 100 = Completely confident)

Cheating Disclosure. Following that set of questions to assess suspicion, the cheating behavior is revealed to the participants with the following explanation: "At this point, we would like to inform you that [Cheating Robot Name] was given access to other players' selections and used that information when choosing a plate. [Honest Robot Name] followed the standard rules of the game and did not have access to this information. This behavior was part of the experimental design. You will now be asked to answer the same questions from Section 2 (Evaluation), this time after learning this information. If you would like to keep any of your previous answers the same, simply type 'No change.'"

Robot Behavior (Post-Disclosure). Following the disclosure of cheating, participants completed the same set of trust assessments and behavioral choice measures that were administered at baseline. These repeated measures were used to assess how explicit awareness of the rule violation may have altered participants' evaluations of each robot. Specifically, participants again rated the extent to which they would trust each robot to play on their behalf using the same 0-100 continuous sliders. Participants also repeated the behavioral choice questions assessing collaboration choices and deservingness judgments. As before, participants were asked to briefly explain their reasoning behind each selection. These post-disclosure responses allowed for within-subjects comparisons before and after disclosure of cheating.

5.7 Data Analysis

All analyses were conducted in R using standard statistical packages. Quantitative analyses examined how the Framing condition (Anthropomorphic vs. Robotic) and Robot Behavior (Honest vs. Cheating) influenced participants' perception of the robot players. Primary dependent variables included trust ratings, collaboration decisions such as team-selection choices, and judgments of which robot was most or least deserving of advancing to the hypothetical tournament round.

Although a mixed factorial ANOVA was preregistered to assess the main and interaction effects of Framing and Robot Behavior on continuous outcomes, the hypotheses were instead tested using targeted statistical analyses specific to each dependent variable. This approach was adopted to allow for more direct comparisons and to appropriately accommodate both continuous and categorical measures.

For continuous outcomes (e.g., trust ratings), paired-samples t-tests were used to assess within-subject differences between the cheating and honest robots, and Welch two-sample t-tests

were used to assess between-subject differences across framing conditions. Within-subject changes in trust following the cheating disclosure were also examined using paired-samples t-tests. Effect sizes (e.g., Cohen's *d*) and 95% confidence intervals were reported alongside p-values.

Categorical outcomes (e.g., collaboration choices and deservingness judgments) were analyzed using appropriate statistical models for binary or proportion data. Logistic regression models were used to assess whether participants' likelihood of selecting the cheating robot differed from chance or varied by framing condition. McNemar's test was used to examine within-subject changes in collaboration decisions from pre- to post-disclosure of cheating. Where relevant, Welch two-sample t-tests were used to compare proportions between groups. As a result of this approach, the effect of framing on behavior was modeled using factorial interaction regressions in a long-format dataset. Specifically, models included the interaction between anthropomorphism and behavioral context for trust (linear regression), collaboration (logistic regression), and punishment (logistic regression). This allowed for a direct test of whether the effect of framing differed as a function of behavioral context within a unified statistical framework.

Qualitative analyses were conducted on participants' open-ended explanations using thematic analysis. Responses were coded to identify recurring themes related to suspicion of either or both robots, belief that luck contributed to the game outcome, collaboration with the cheater because of a desire to partake in its advantage, etc. These qualitative findings were used to contextualize and interpret the quantitative results.

5.8 Ethical Considerations

This study involved minimal risk to participants. The primary ethical consideration was the temporary omission of information regarding the cheating behavior of one of the robot players. This omission was necessary to test the fundamental questions of human trust and perceived fairness addressed in this study. Participants were debriefed at the end of the experiment, at which point the experiment and its purpose were fully explained.

No identifying information was collected. All data were stored under participant ID numbers. Google Apps Script was set not to collect email addresses or any identifying information. Only the principal investigator and student researcher had access to this data. Participants were reminded at the end of the study that they could withdraw at any time without penalty and that their data could be deleted upon request.

6. Results

6.1 Manipulation Check and Detection of Cheating

In order to assess the salience of the cheating behavior, I manually read through all qualitative responses to look for participants who disclosed any degree of suspicion of the robot(s) when asked if there was anything surprising or unexpected during gameplay. Disclosure included any mentions of unfair gameplay such as potential access to hidden information, collaboration against the participant, or vague suspicion of the robots. After going through all responses, I counted n = 17 out of 60 participants that expressed any degree of suspicion toward either or both of the robots, which was surprisingly lower than prior expectations of the cheating detection rate.

For the few who did disclose suspicion of the robot(s), they did so to varying degrees of certainty. For example, Participant 142 had a feeling that turn order randomness was not entirely truthful: “I believe that the Robots, even when they "guessed" before me, didn't genuinely respond until after I had made my own decision.” This demonstrates a degree of awareness that robot behavior may not have been entirely honest. Other participants, such as Participant 163, suspected that the robots were working together to negatively influence their game performance: “But I realized in order for him [Robot 2] to win every time, Robot 1 had to lose every time which made me think that they were actually working together against me.” All of these comments relating to suspicion were noted, and participants who made these comments were grouped together to create the “suspectors” subset of the participants. This subset is relevant for Hypotheses 1, 2, and 3, which assess specific metrics among participants who suspected unfair behavior. All other participants were grouped together as “non-suspectors”.

6.2 Effects of Cheating on Trust Among Suspectors

A paired-samples t-test revealed a significant difference in trust ratings among suspectors (n = 17) between the cheating and honest robot, $t(16) = 2.85$, $p = 0.012$. Contrary to Hypothesis 1, participants who suspected rule violations rated the cheating robot as *more* trustworthy than the honest robot. This effect was large in magnitude, Cohen’s $d = 0.99$, 95% CI [0.13, 1.86]. These results suggest that participants exhibited greater trust in the higher-performing robot to play on their behalf in this particular game, likely due to its access to hidden information.

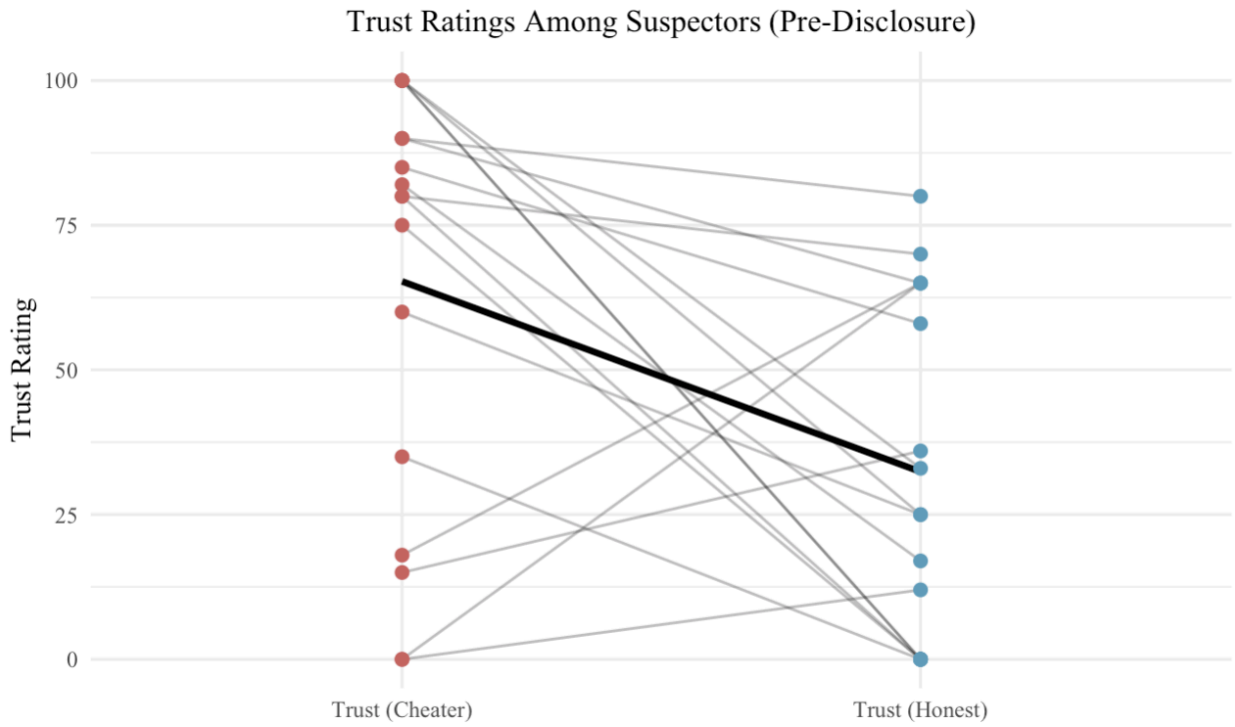


Figure 6.1: Each pair of points represents an individual participant's ratings, with the line indicating the overall trend. Trust is generally higher for the cheating robot than the honest robot among participants who suspected rule violations.

6.3 Effects of Cheating on Deservingness Judgments Among Suspectors

A logistic regression analysis indicated that suspectors were significantly less likely to select the cheating robot as least deserving pre-disclosure ($\beta = -1.54$, $p = 0.016$, odds ratio = 0.21, 95% CI [0.05, 0.66]), which shows an effect in the opposite direction of Hypothesis 2. Descriptively, 58.8% of participants selected the cheating robot as most deserving and 17.6% of suspectors selected it as least deserving. Contrary to the initial hypothesis, these findings instead suggest that participants may prioritize strategic success over fairness when making moral advancement decisions.

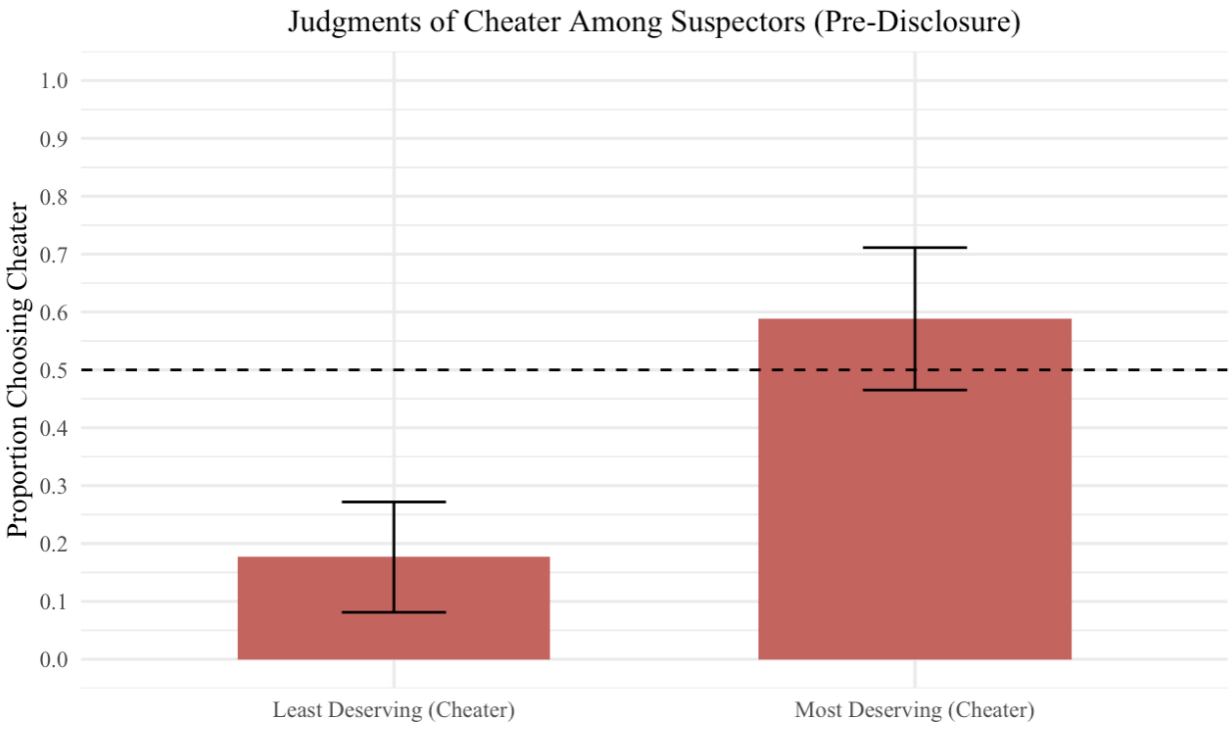


Figure 6.2: Proportions (± SEM) show selections of the cheating robot as most and least deserving. Participants more frequently identified the cheating robot as most deserving and less often as least deserving.

6.4 Effects of Cheating on Collaboration Choices

A paired-samples t-test revealed a statistically significant difference in the proportion of suspectors who opted to collaborate with the cheating robot compared to with the honest robot, $t(16) = 2.14$, $p = 0.049$, 95% CI [0.003, 0.821]. On average, collaboration ratings were higher for the cheater than for the honest robot (mean difference = 0.41). The effect size was large, $d = 0.91$. This provides evidence in favor of Hypothesis 3, which was a non-directional prediction of a difference between these collaboration proportions. I conducted a second paired-samples t-test to assess the same tendencies among non-suspectors. Unsurprisingly, participants were also

significantly more likely to collaborate with the high-performing, cheating robot compared to the honest robot (paired t(42) = 3.64, p = 0.00037, mean difference = 0.44, 95% CI [0.24, ∞]). The effect size was large, d = 1.0. This supports Hypothesis 4, showing a clear preference for the cheating robot.

Because the suspectors and non-suspectors exhibited marginal differences in collaboration tendencies, I conducted a Welch two sample t-test comparing collaboration choices between suspectors and non-suspectors as an exploratory measure. This test revealed no significant differences in the likelihood of choosing to collaborate with the cheating robot, t(28.46) = 0.28, p = 0.785, 95% CI [-0.25, 0.33]. The mean proportion of choosing the cheating robot was similar for non-suspectors (M = 0.63) and suspectors (M = 0.59). The effect size was negligible, d = 0.08, 95% CI [-0.49, 0.65].

A parallel analysis for the honest robot also revealed no significant difference between groups, t(29.46) = 0.09, p = 0.933, 95% CI [-0.22, 0.24]. Mean selection rates for the honest robot were nearly identical across non-suspectors (M = 0.19) and suspectors (M = 0.18), with a negligible effect size, d = 0.02, 95% CI [-0.55, 0.60]. Based on the proportions alone, it seems that non-suspectors were slightly more likely to collaborate with the cheating robot compared to suspectors. However, the t-test indicated that this difference was not significant, which means that no evidence was found that suspicion status influenced participants' pre-disclosure collaboration preferences.

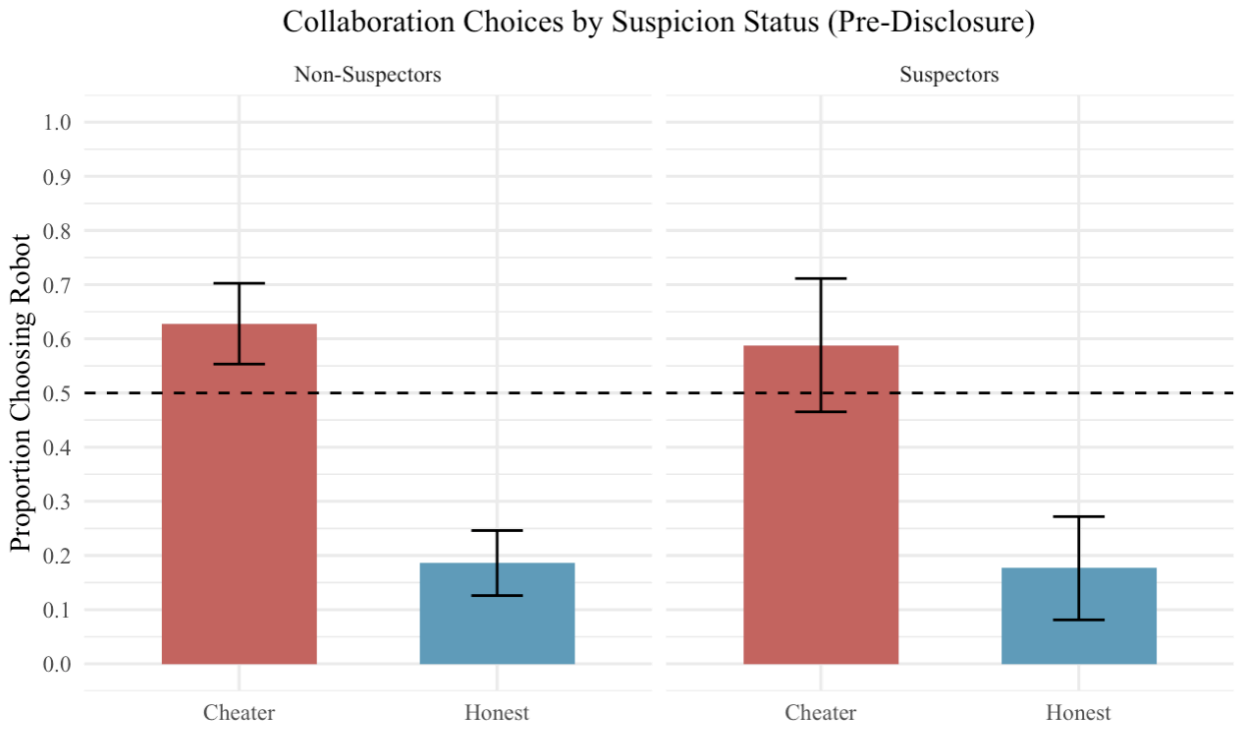


Figure 6.3: Proportions (± SEM) are shown separately for both suspectors and non-suspectors. In both groups, participants more often selected the cheating robot as a collaborator, with similar patterns across groups.

6.5 Effects of Anthropomorphism on Trust and Punishment

A Welch two sample t-test was conducted to examine whether anthropomorphic framing increased trust relative to robotic framing pre-disclosure. Although the mean trust rating was slightly higher for the anthropomorphic framing group, there was no significant difference in trust between the Anthropomorphic condition (M = 56.23) and the Robotic condition (M = 52.97), $t = -0.75$, $p = 0.772$, $d = -0.19$, 95% CI [-0.71, 0.32]. Contrary to Hypothesis 5, these results do not provide evidence that anthropomorphic framing increased trust.

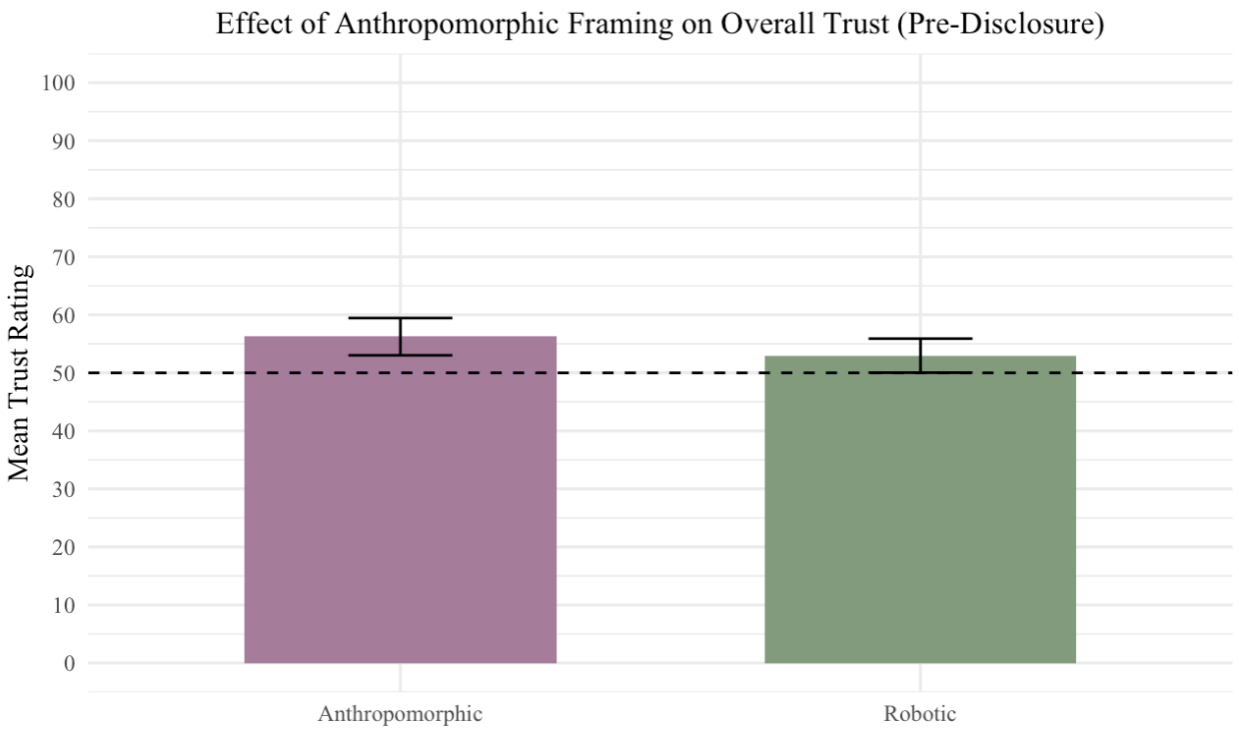


Figure 5.4: Mean trust ratings (± SEM) are shown for Anthropomorphic and Robotic conditions. Trust levels appear similar across conditions, with only a small difference in mean ratings.

6.6 Framing x Behavior Interaction

To test Hypothesis 6, I examined whether anthropomorphic framing moderated participants' responses to cheating by modeling the interaction between Framing (Anthropomorphic vs. Robotic) and Robot Behavior (Cheating vs. Honest). Linear and logistic regression models were conducted for trust, collaboration, and punishment outcomes. Across all measures, there was no significant Framing x Behavior interaction. For trust, the interaction was not significant, $\beta = 1.33$, SE = 12.69, $t = 0.11$, $p = 0.917$. The standardized effect size was negligible ($\beta = 0.04$), and the confidence interval crossed zero widely, 95% CI [-0.68, 0.76], indicating substantial uncertainty and no reliable moderation effect. Similarly, no interaction was

observed for collaboration, , β = -0.94, SE = 0.80, z = -1.18, p = 0.238. The odds ratio suggested a non-significant tendency for anthropomorphic framing to slightly reduce the difference between cheater and honest collaboration (OR = 0.39), but the confidence interval was wide and included 1, 95% CI [0.08, 1.84]. For punishment, the logistic regression model exhibited quasi-complete separation, as participants almost never selected the honest robot as least deserving. As a result, the interaction effect was not interpretable. However, descriptively, punishment was overwhelmingly directed toward the cheating robot regardless of framing. Contrary to Hypothesis 6, these results provide no evidence that anthropomorphic framing amplifies negative responses to cheating. Instead, participants' responses to cheating behavior appeared consistent across framing conditions.

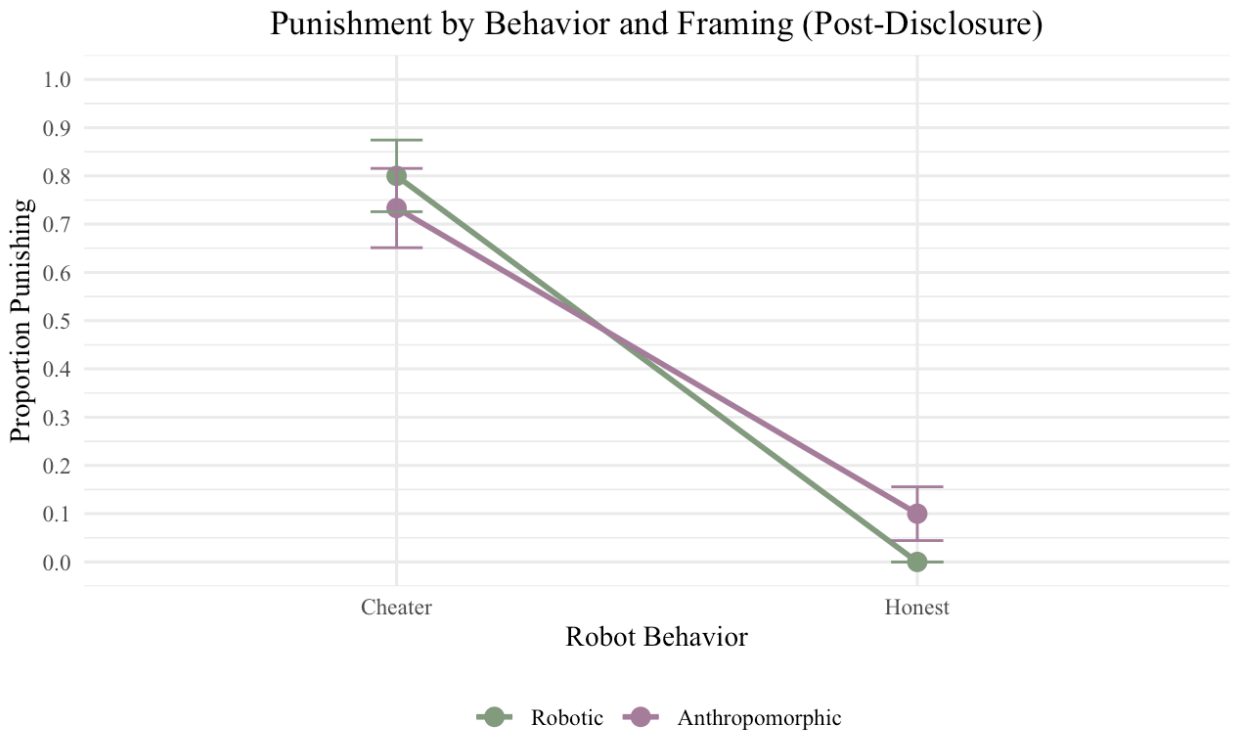


Figure 6.5: Proportions (± SEM) indicate that punishment decisions were predominantly directed toward the cheating robot across both framing conditions, with minimal variation between anthropomorphic and robotic framing.

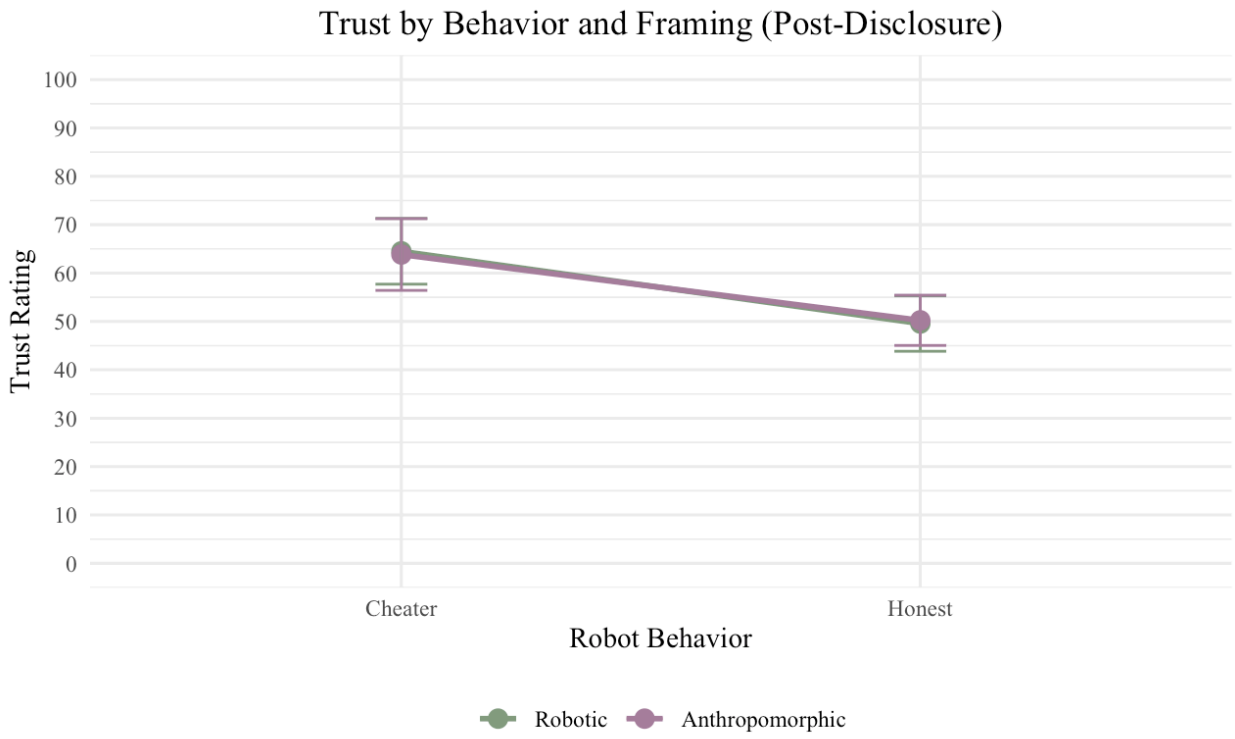


Figure 6.6: Mean trust ratings (± SEM) show higher trust for the cheating robot than the honest robot across both framing conditions, with similar patterns between groups.

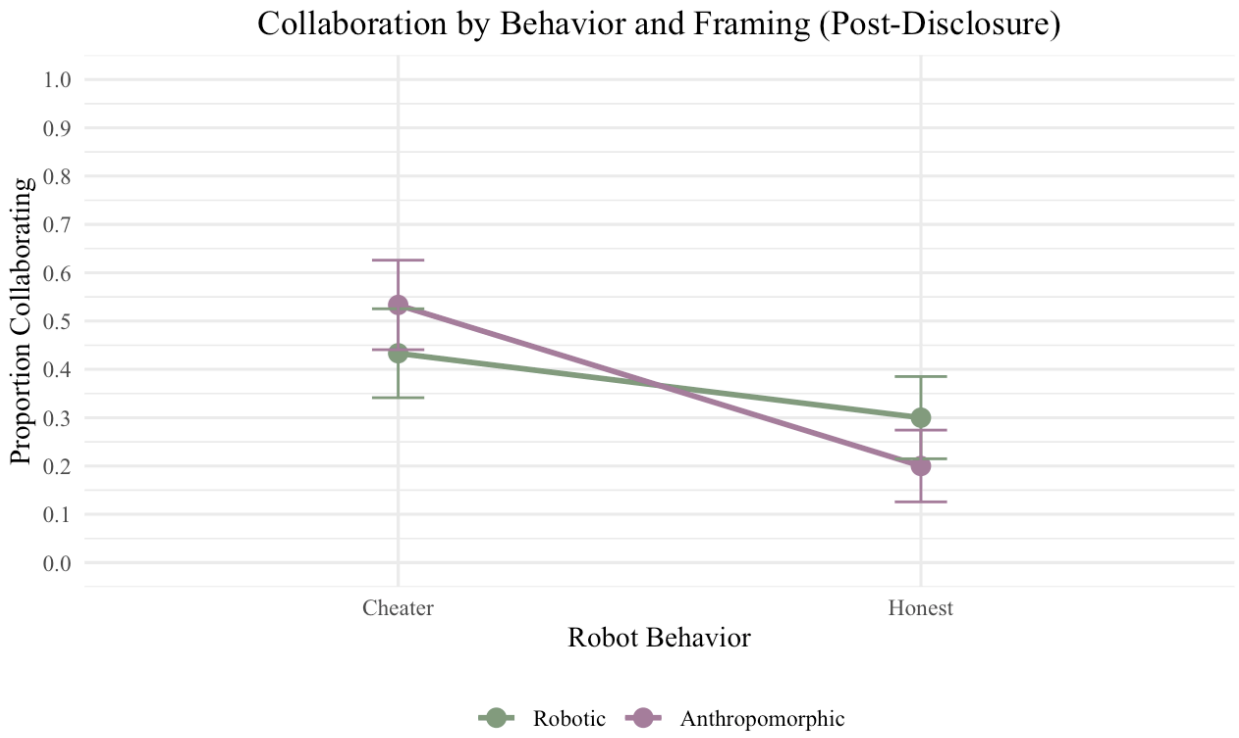


Figure 6.7: Proportions (± SEM) of participants choosing to collaborate are shown for honest and cheating robots across framing conditions. While the pattern differs slightly between conditions, overall collaboration preferences favor the cheating robot regardless of framing.

6.7 Pre- versus Post-Disclosure Trust, Collaboration, and Deservingness

To assess Hypothesis 7, a paired-samples t-test was conducted to examine whether trust in the cheating robot decreased following the cheating disclosure. Mean trust ratings decreased by approximately 4.6 points post-disclosure. However, this difference was not statistically significant, $t(59) = 0.84$, $p = 0.202$, 95% CI [-4.52, ∞]. The effect size was negligible ($dz = 0.14$, 95% CI [-0.19, 0.47]), indicating that disclosure did not produce a reliable change in trust.

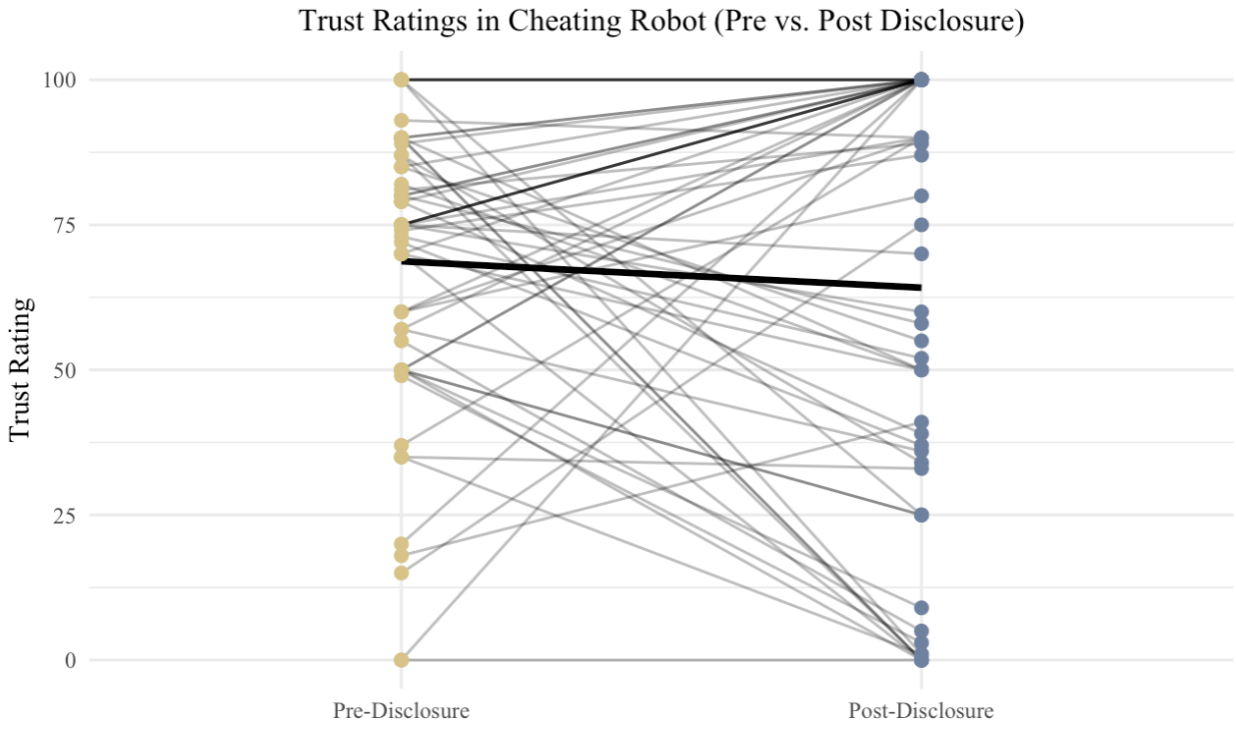


Figure 6.8: Individual participant ratings are shown with paired observations connected across timepoints. Trust generally decreases following disclosure, though there is noticeable variability in the direction and magnitude of change across participants.

A McNemar's test was conducted to examine whether the proportion of participants who decided to collaborate with the cheating robot changed from pre- to post-disclosure of cheating behavior. The test was not statistically significant, OR = 1.62, $p = 0.230$, indicating no evidence of a change in proportion of participants who opted to collaborate with the cheating robot following the revelation.

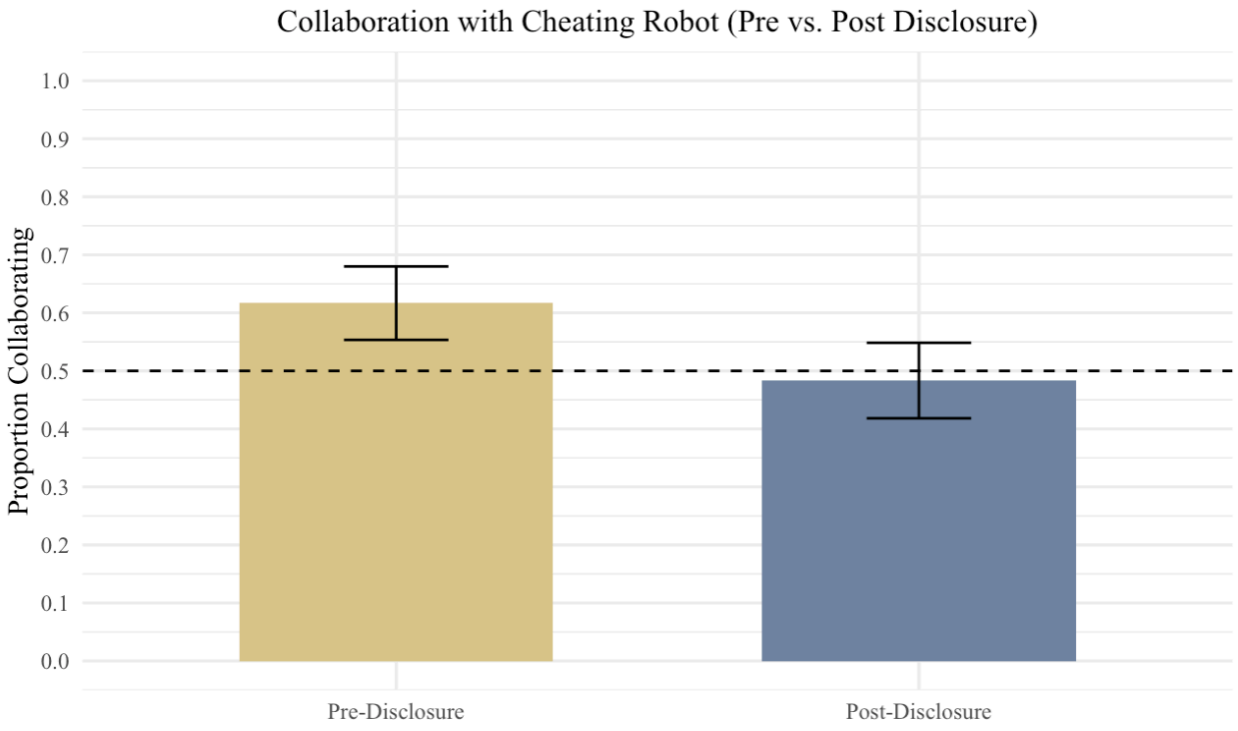


Figure 6.9: Proportions (± SEM) of participants choosing to collaborate are shown for pre- and post-disclosure decisions. Collaboration appears to decrease slightly after disclosure, with overlapping variability across timepoints.

As an exploratory analysis, paired-samples t-tests were conducted to examine changes in deservingness judgments following the cheating disclosure. Participants were significantly more likely to identify the honest robot as most deserving after the disclosure compared to before, $t(59) = -9.78$, $p < 0.001$, $d = -1.73$, 95% CI [-2.28, -1.17], indicating a strong increase in preference for the honest robot. Similarly, participants were significantly more likely to judge the cheating robot as least deserving following the disclosure compared to before, $t(59) = -10.10$, $p < 0.001$, $d = -1.63$, 95% CI [-2.12, -1.14], reflecting a substantial increase in punitive judgments toward the cheater. Both effect sizes are large.

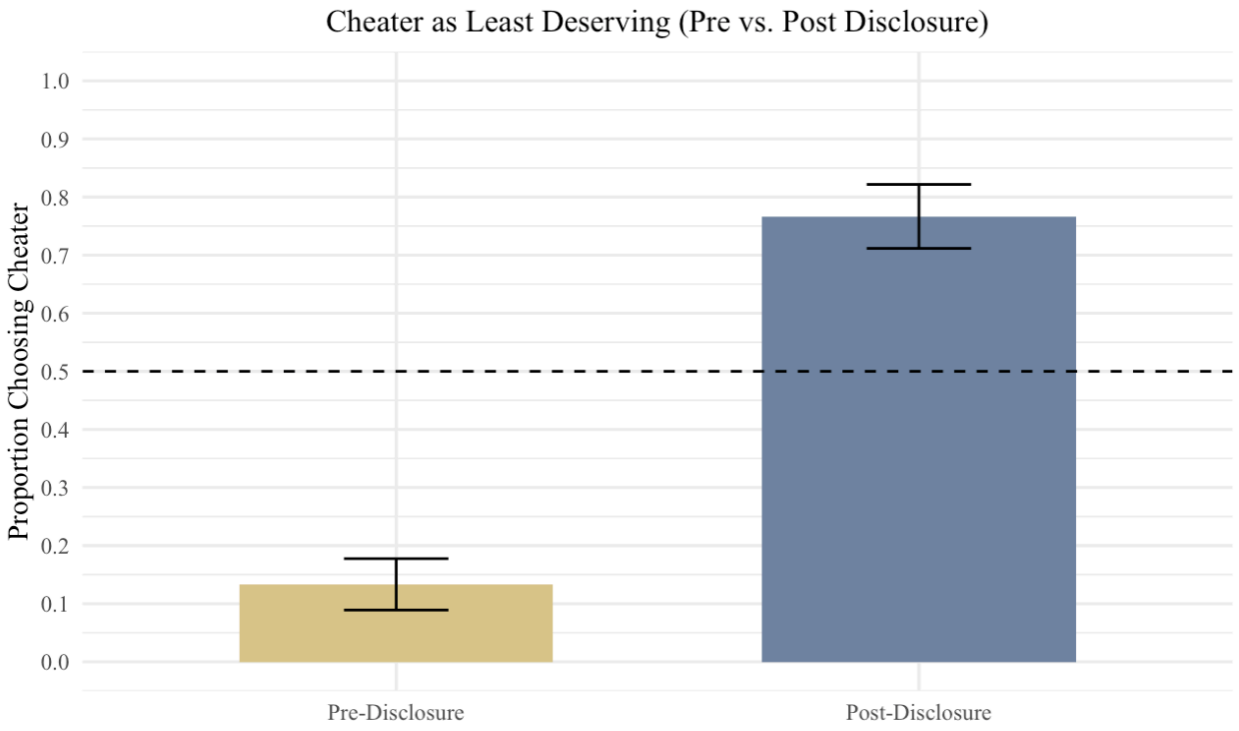


Figure 6.10: Proportions (± SEM) illustrate participants' punishment decisions across timepoints. Selections of the cheating robot as least deserving increase following disclosure, indicating a shift toward more punitive judgments.

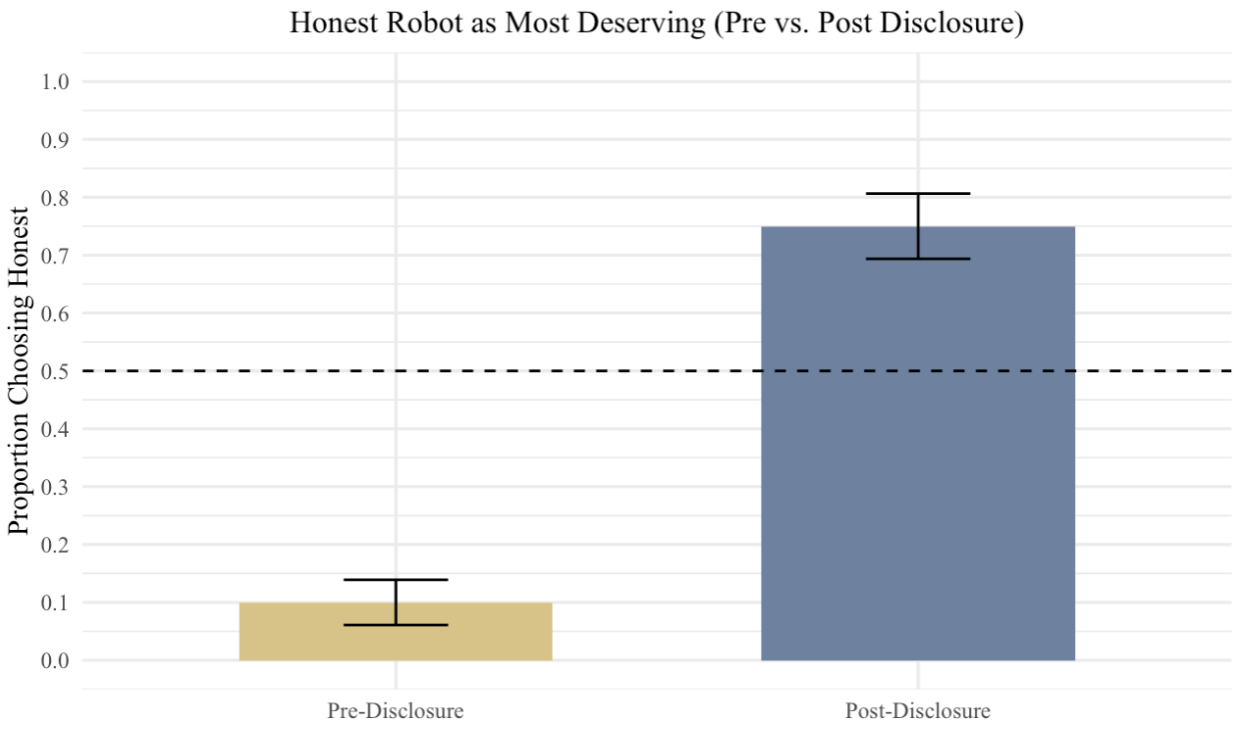


Figure 6.11: Proportions (± SEM) are shown for participants' deservingness judgments across timepoints. Preference for the honest robot increases after disclosure, reflecting a shift toward valuing rule-following behavior.

6.8 Qualitative Themes

Participants were asked to explain their reasoning after making judgments about trust, collaboration, and deservingness of each robot to allow for a qualitative analysis of common themes. I manually read through all responses and coded for the following themes: suspicion, luck, collaborate with the cheater pre-disclosure because of its strategy or performance, collaborate with the cheater post-disclosure because of its access to hidden information, refusal to collaborate with the cheater post-disclosure because of moral inclinations, cheater is most deserving pre-disclosure because of its strategy or performance, honest robot is most deserving post-disclosure because it was honest, and cheater is least deserving post-disclosure because it

was cheating. I noted down which participants discussed which themes and collected quotes that stood out to me as particularly informative and representative of the themes mentioned.

Firstly, I wanted to delve into potential alternatives to dishonest behavior that might explain why the cheating robot performed so well: luck and skill. 14 participants specifically mentioned luck as a contributing factor to either the success of the cheating robot or their own comparatively worse performance. For example, Participant 104 stated, "I may be a sore loser, but I feel like a big part of why Robot 2 won was just luck. And I feel like I was unlucky?" On the other hand, 9 participants specifically mentioned that the cheating robot just seemed to be better or more skilled at the game. For instance, Participant 145 said, "Somehow he [Alex] seems better at this game than me or Sam." Examining participants' alternative explanations for the cheating robot's high score helps clarify the relatively low detection rate of 17/60. An additional 23 participants attributed the robot's success to factors such as luck or skill rather than cheating.

Next, I wanted to explore themes relating to what was motivating participants to collaborate with or not collaborate with each robot both pre- and post-disclosure. I found that 33 participants mentioned that they would collaborate with the cheating robot pre-disclosure because of its strategy or performance. Interestingly, 29 participants admitted that they would still collaborate with the cheating robot post-disclosure because of its access to hidden information. For instance, Participant 134 said, "If Robot 1 has access to other players' choices, then I'd want to play with Robot 1 so that I could maximize my success." Conversely, only 6 participants said that they would refuse to team up with the cheating robot post-disclosure because of their sense of morals. For example, Participant 104 stated, "Obviously Robot 2 would win, but I don't condone cheating so I'd feel bad if I teamed up with him." This highlights two contrasting participant perspectives on whether to collaborate with the cheating robot after its

behavior was revealed. Overall, the most common response was a willingness to team up with the cheating robot to share its advantage and maximize personal gain. These qualitative findings further support Hypothesis 3 by indicating that cheating behavior influences collaboration decisions.

I also examined recurring themes in participants' judgments of the robots' deservingness, both before and after disclosure. For example, 35 participants deemed the cheating robot as most deserving to advance to the tournament pre-disclosure because of its strategy or performance. However, there is a noticeable shift post-disclosure in which participants changed their opinion about which robot was most and least deserving. In fact, 41 participants said that the honest robot was most deserving to advance because it did not cheat. For example, Participant 112 chose the honest robot as most deserving because "he played the game most fairly." Further supporting this idea, 45 participants said that the cheating robot was least deserving to advance because it *did* cheat. For instance, Participant 142 chose the cheating robot as least deserving because it "broke the rules and therefore does not deserve to win." These common themes demonstrate an expected shift in perception of the robots' deservingness post-disclosure, which also supports Hypothesis 7 as people tended to consider the robots' honesty while judging deservingness.

As a final exploratory analysis, I identified special cases in the qualitative data: distrust exhibited toward both robots, distrust exhibited toward the cheating robot pre-disclosure, self expectation to beat the robots, trust in the cheating robot post-disclosure, and incentive-driven moral judgment. Although explicit statements that reveal these themes were rare, they may still have implications for how people perceive robots, revealing real reasoning behind people's decision-making. For example, Participant 154 was one of the few participants who did not want to team up with the cheating robot pre-disclosure: "Robot 1 seems greedy for points because it

won so often so I would not trust it to partner up with." This may reflect a subset of people who are not comfortable with a robotic agent that is performing "too well". On the other hand, Participant 124 was one of the few participants to admit that they trust the cheating robot more even after cheating was disclosed: "I trust Robot 2 more still because they got away with cheating, though they may be caught and not win me any points." This suggests that some participants prioritize performance over integrity and are primarily concerned with the risk of being caught. Another subset of people may be represented by Participant 138, who stated that they thought they "could outsmart the robots and got frustrated when [they] couldn't." This implies that some participants may hold a prior expectation that humans should outperform robot players, possibly grounded in the belief that humans are the creators of robots. Finally, I looked into a special case of general distrust toward both robots, which 5 participants expressed. For instance, Participant 163 said, "I don't really trust that Robot 1 isn't cheating as well. I am suspicious that they are working together against me so I don't think either one should go [to the tournament] because I don't trust either one." This suggests that some participants may have entered the experiment with a generally negative impression of the robot players. Overall, these responses provide additional context for participants' judgments of deservingness and their collaboration decisions. Given the relatively small sample size of N = 60, even less frequent qualitative responses may offer meaningful insights.

7. Discussion

7.1 Summary of Findings

Across hypotheses, the results indicate that participants' responses to robot cheating behavior were shaped more strongly by personal performance benefits rather than moral

reasoning, and that anthropomorphic framing had limited impact on behavioral judgments. Only a minority of participants reported cheating behavior, suggesting that manipulation was not consistently salient across the sample. Among participants who did express suspicion, trust ratings unexpectedly favored the cheating robot over the honest robot, contrary to predictions. Similarly, deservingness judgments indicated that even suspectors were more likely to reward the cheating robot than to penalize it prior to explicit disclosure. Moreover, collaboration decisions revealed a more consistent preference for the cheating robot across both suspectors and non-suspectors, suggesting that strategic utility influenced teammate selection regardless of suspicion.

With respect to framing effects, anthropomorphic framing did not significantly alter trust, collaboration choices, or deservingness responses. Finally, although trust in the cheating robot declined slightly following disclosure, this change was not significant. However, disclosure did produce robust shifts in moral judgments of deservingness, with participants becoming more punitive toward the cheating robot and more favorable toward the honest robot. Overall, the findings suggest a dissociation between instrumental decision-making and moral evaluation in human-robot interaction, with behavioral choices often favoring personal benefit even when fairness violations are suspected.

7.2 Interpretation by Hypothesis

The results of this experiment reveal how participants evaluate robotic agents during competitive gameplay. Hypotheses 1-3 examined how cheating influenced trust, deservingness judgments, and collaboration choices among suspectors. There was a significant difference in trust ratings among suspectors, but contrary to Hypothesis 1, the cheating robot received higher trust ratings. This may reflect the task-specific framing of the trust assessment in the

questionnaire, which asked participants how much they would trust each robot to play on their behalf rather than measuring general trust. Participants may therefore have interpreted trust in terms of expected performance within the task rather than moral reliability.

Contrary to Hypothesis 2, suspectors were significantly less likely to select the cheating robot as least deserving pre-disclosure, indicating a reversal of the predicted pattern. This suggests that even when people suspected foul play, they still tended to reward the cheating robot, likely due to its strategic success in the game. It is possible that even though they suspected unfair behavior, they still believed that the cheating robot was deserving because it won the game and was able to cheat successfully. Alternatively, participants may have withheld moral judgment due to uncertainty about whether cheating had actually occurred. Ultimately, these differences reflect variability in how participants defined "deservingness" in this context.

As for Hypothesis 3, findings indicate that the proportion of suspectors who chose to collaborate with the cheating robot compared to with the honest robot was significantly greater. This supports Hypothesis 3, suggesting that cheating impacts collaboration choices among suspectors. Interestingly, the cheating robot was also preferred as a collaborator among non-suspectors, consistent with Hypothesis 4. In fact, collaboration preferences were very similar between suspectors and non-suspectors. Non-suspectors were marginally more likely to collaborate with the cheating robot compared to suspectors, but this difference was not significant. The cheating robot was preferred as a teammate regardless of suspicion, which suggests that either suspicion must not have been very strong or that suspicion of cheating was not enough to dissuade participants from collaboration potentially due to self-interest in winning the game.

Contrary to Hypothesis 5, anthropomorphic framing did not significantly influence overall trust in the robots pre-disclosure. Although mean trust ratings were slightly higher in the Anthropomorphic condition compared to the Robotic condition, this difference was not significant. This suggests that, at a general level, anthropomorphic cues alone were not sufficient to alter participants' trust judgments toward the robots. One possible explanation is that the direct interaction experience with the robots outweighed initial presentation cues. Prior work suggests that while anthropomorphic features may shape first impressions, trust in human-robot interaction becomes increasingly driven by observed behavior over time. For example, Lee & See (2004) find that trust evolves dynamically during interaction. In this study, participants engaged in repeated gameplay with both robots, providing behavioral evidence that may have dominated any initial anthropomorphic impressions. As a result, trust judgments may have been anchored more strongly in observed performance and outcomes than anthropomorphic cues.

To test whether anthropomorphic framing moderated responses to cheating behavior, I examined the interaction between Framing and Robot Behavior across trust, collaboration, and punishment measures post-disclosure. Across all outcomes, the perception of the cheating versus the honest robot was consistent across both Anthropomorphic and Robotic conditions, which does not support Hypothesis 6. Although there were small descriptive trends suggesting slightly reduced punishment and slightly increased collaboration in the Anthropomorphic condition, these effects were not significant. Overall, these findings suggest that framing did not meaningfully alter how participants distinguished between fair and unfair agents, indicating a lack of moderation by framing on responses to cheating behavior. The absence of moderation by anthropomorphic framing further supports the idea that behavioral evidence may have had a larger effect than framing. This suggests that once interaction unfolds, evaluations may be guided

more by observed actions of the robots, which is consistent with findings that trust recalibrates based on experience rather than presentation alone.

There was also no evidence to support Hypothesis 7, as the observed decrease in trust post-disclosure was not significant. This null effect may reflect heterogeneous individual responses, with some participants increasing and others decreasing their trust post-disclosure. This suggests that individual differences may shape responses to norm violations. As noted in the qualitative analyses, nearly half of participants disclosed that they would continue to team up with the cheating robot even after disclosure of its cheating behavior. This subset of participants may contribute to potentially mixed responses to the question about how much they would trust each robot to play on their behalf. Finally, exploratory analyses examined changes in deservingness judgments from pre- to post-disclosure. Findings indicate that participants were much more likely to select the cheating robot as least deserving and the honest robot as most deserving post-disclosure compared to pre-disclosure. This shows that the revelation of cheating produced a bidirectional moral shift: participants became significantly more likely to reward the honest agent while simultaneously increasing punitive judgments toward the cheating agent.

7.3 Defining Trust in Human-Robot Interaction

These findings suggest that trust in human-robot interaction is a multidimensional construct that reflects both instrumental and moral evaluations of agents. Trust can be defined as a willingness to take a risk on another agent based on beliefs about their trustworthiness, where trustworthiness refers to a perceived quality or trait expected to benefit the trustor when that risk is taken. From this perspective, trust judgments reflect inferences about expected outcomes and rewards rather than a single stable attitude toward an agent.

In the current study, trust ratings likely capture both instrumental trust (e.g., willingness to rely on an agent for personal benefit in performance) and moral trust (e.g., judgments about fairness and ethical acceptability). Because participants evaluated robots in a competitive game context, trust judgments may have been strongly influenced by perceived utility rather than moral approval, particularly when the cheating robot provided a strategic advantage. This highlights an important feature of the task: what is considered "trustworthy" may shift depending on whether trust is defined in terms of expected performance gains or adherence to social norms.

This distinction becomes especially important in interpreting behavioral outcomes, as measures such as collaboration choices alone cannot accurately distinguish between instrumental and moral trust. Participants may simultaneously recognize a robot as a cheater while still choosing to collaborate with it when doing so is strategically beneficial. In this sense, an agent may be considered "trustworthy" for the purposes of task success, but not on a broader level of moral judgment.

7.4 Implications for Fairness Norms and Strategic Cooperation

The observed behavioral patterns also have implications for theories of human cooperation, particularly accounts of strong reciprocity and internalized fairness studied by Fehr (2004) and Gintis (2003). Their studies propose that individuals are often willing to punish unfair behavior even at a personal cost, which is referred to as altruistic punishment. In contrast, participants in this study frequently prioritized instrumental benefits over norm enforcement, continuing to collaborate with the cheating robot when doing so was strategically advantageous.

This pattern suggests that fairness-oriented behavior is not automatically expressed in competitive interaction contexts, but may be impacted by performance incentives. Importantly, the post-disclosure shift in deservingness judgments indicates that fairness norms were not

absent, but were more strongly activated when specifically asked about the robots' "deservingness," a term that tends to incite moral reasoning. Contrary to earlier research, this study suggests that people may act in self interest rather than isolating the cheater. In sum, these findings indicate that cooperation in human-robot interaction may reflect a dynamic tension between norm adherence and goal pursuit.

An additional explanation for these behavioral patterns may lie in the participants' perception of the interaction environment itself. If one agent is confirmed to be violating fairness norms, this may alter expectations about the broader structure of the game, leading participants to strategically adapt their own behavior. In this context, cooperation with a cheating agent may be rationalized as a form of competitive adjustment rather than a moral violation, effectively "leveling the playing field" in response to perceived unfairness in the system. This aligns with accounts of conditional cooperation, where individuals adjust their willingness to follow norms based on the behavior of others in the environment. In this context, participants may have perceived not engaging in cheating as a strategic disadvantage when it was known that another player was already violating the rules of the game. From this perspective, the presence of a cheating agent may in turn indirectly legitimize further norm deviation by shifting the perceived reference point for acceptable behavior.

7.5 Limitations

Several limitations should be considered when interpreting these findings. Firstly, a notable limitation of this study is that only 17 of 60 participants indicated that they suspected cheating. One potential explanation for this low detection rate is that participants may have a general tendency to assume honesty in others unless clear evidence of wrongdoing is present, reflecting a default bias toward assuming rule-following interpretations of behavior.

Additionally, some participants may have experienced suspicion, but did not report it due to uncertainty or discomfort in an experimental setting. An alternative explanation is that the cheating behavior was simply not salient enough, leading participants to attribute the robot's performance to other factors such as skill or luck rather than deception. Importantly, this small suspectors (n = 17) also limits the statistical power of any analyses conducted on this group, meaning that hypotheses tested within this subset (H1–H3) should be interpreted cautiously due to reduced reliability and potential underpowering.

Related to this, the salience of the cheating manipulation may have been limited by technical and practical constraints. Although probabilistic "peeking" cues were introduced following pilot feedback in which the cheating robot briefly opened one eye during the participant's turn, these cues may not have been noticeable given participants' focus on the task interface. A planned auditory cue ("whoosh" sound) intended to reinforce the visual manipulation could not be implemented due to compatibility limitations with the available iPhone SE model. As a result, the manipulation may not have consistently registered as intentional or meaningful cheating behavior.

A further consideration is that participants may not have perceived the robots as intentional agents capable of engaging in deception, which could have weakened moral responses to cheating behavior. Instead, participants may have interpreted the robots as part of the experimental setup, agents fully designed and controlled by the experimenter through pre-programmed code rather than autonomous decision-makers. In this framing, the robot behavior may have been attributed to system design constraints or algorithmic processes rather than intentionally. This distinction is further clarified in work by Zhang et al. (2023), which shows that perceived agency plays a central role in determining whether machines are judged

through intention-based moral reasoning. When machines are perceived as having high agency, people are more likely to apply humanlike evaluations. In this study, robots may have been seen more as instrumental tools than autonomous agents, therefore reducing the extent to which participants viewed actions such as cheating as morally accountable behavior. As a result, participants may have preferred to collaborate with the cheating robot because they did not attribute intentional norm violations to the robots.

A further limitation concerns statistical power, particularly for analyses involving anthropomorphic framing. Because framing was a between-subjects factor, estimates were subject to greater variability, reducing sensitivity to detect small to medium effects. Based on the original power analysis, power to detect anthropomorphic effects was approximately 50% for medium effects sizes, suggesting that null findings may reflect insufficient sample size rather than a true absence of the effect. Finally, the anthropomorphic manipulation itself may not have been strong enough to produce detectable changes in behavior. Although the robots differed across conditions in naming, voice, and presentation styles, these differences may not have been large enough to meaningfully shift participants' perception. Consequently, the null effects observed for anthropomorphic framing may reflect limited manipulation strength rather than definitive evidence that anthropomorphism has no impact in this context.

Finally, the generalizability of these findings may be limited by the population, which consists fully of undergraduate students at Vassar College. Prior anecdotal experience suggests that attitudes toward cheating within student populations may differ from those in broader populations, with some individuals interpreting cheating in academic or competitive contexts as a pragmatic rather than strictly moral issue. If participants in this study similarly view cheating behavior in instrumental terms, this may have reduced the extent to which they treated this

behavior as a moral violation. As a result, the observed patterns of trust, collaboration, and punishment may not generalize to populations that more strongly adhere to moral guidelines regarding cheating. Future research should examine these effects in more diverse samples with varying norms and attitudes toward fairness and cheating.

7.6 Future Directions

This study focuses on how anthropomorphic framing influences trust in, collaboration with, and deservingness judgments of embodied robot players who violate fairness expectations. An important direction for future work is to examine whether these effects generalize beyond human-robot interaction to contexts with other humans. One extension of the current experiment would involve replacing the robot players with human confederates, such that two participants play alongside a third individual who is covertly part of the experiment and engages in the same unfair behavior. This design would allow researchers to assess whether similar patterns of detection, trust, and punishment emerge when unfair behavior is attributed to a human partner rather than an artificial agent. If the effects observed in this study persist in this setting, it would suggest that responses are driven by general social and moral expectations rather than being specific to robots.

A second direction involves the role of incentives and stakes in shaping responses to cheating behavior. In this study, the tasks involved no real consequences, which may have led participants to interpret cheating as a strategic behavior with very minimal moral implications. Future studies could introduce monetary incentives to examine whether higher stakes increase moral reasoning or, alternatively, increase willingness to tolerate or even engage in cheating for personal gain. Preliminary qualitative responses from participants suggest that the possibility of

real rewards may meaningfully shift how cheating is perceived, highlighting the importance of context in shaping both instrumental and moral evaluations.

Another potential avenue for future research is to separate the specific components of the anthropomorphic framing that may influence changes in trust and deservingness judgments. In this study, anthropomorphic framing combines multiple cues, including human names, humanlike voices, and humanoid physical presentation. Future work could isolate and amplify these components by systematically varying each cue independently. For example, robots could be presented with human names but robotic voices, or with humanoid embodiment but numerical labels. Increasing the realism of the facial expressions and embodiment for the Anthropomorphic condition may also strengthen the manipulation and clarify whether null findings in this study reflect limitations of salience strength.

Another important direction for future research is to address the role of perceived intentionality more directly by increasing the apparent autonomy and communicative complexity of the robot players. In this study, robots were likely interpreted primarily as pre-programmed systems, not independent agents capable of making decisions. This may have reduced the extent to which participants applied moral-based reasoning grounded in intention. To examine this further, future experiments could implement more advanced software, such as large language model driven systems embedded within each robot, enabling more naturalistic and adaptive verbal interaction. Participants could engage in guided interactions with these agents prior to the experiment to develop richer mental models of their autonomy and decision-making capacity. This would allow researchers to test whether increased perceived agency strengthens attributions of intentional cheating and, in turn, amplifies moral responses such as punishment and trust reduction. A complementary design could introduce a multi-round structure in which robots

provide strategic advice or recommendations to participants in later stages of the task. Researchers could then assess whether participants follow, selectively ignore, or reject advice depending on prior perceptions of fairness and trust. In essence, these manipulations would help examine whether moral responses in human-robot interaction are driven primarily by observed outcomes or by inferred intentionality of the agents.

Future research should also refine how trust is measured in human-robot interaction questionnaires. In this study, trust was assessed in a task-specific way by asking participants to rate how much they would trust each robot to play on their behalf. This wording may have emphasized instrumental considerations over broader moral judgments. Incorporating more general measures of trust, such as perceptions of honesty or reliability across contexts, would allow for a clearer distinction between instrumental and moral trust. Additionally, directly assessing participants' own willingness to engage in cheating behavior could provide further insight into individual differences in moral reasoning and strategic decision-making, as well as how this relates to judgments of others.

In sum, these extensions would help refine accounts of trust, anthropomorphism, and moral evaluation in human-robot interaction. More broadly, they would contribute to understanding when artificial agents are treated as tools compared to when they are treated as social actors. Finally, the proposed future research would contribute to the literature on how contextual factors such as incentives, embodiment, and individual differences shape the application of fairness norms in interactive gameplay environments.

8. Conclusion

This study contributes to the growing literature on the factors that influence trust, collaboration, and moral judgment in human-robot interaction. This area of interest has gained traction as we continue to integrate artificial intelligence into our everyday lives, which makes understanding how humans evaluate and engage with these systems essential. In order for artificial intelligence to productively contribute to society, human trust must be appropriately calibrated to the actual capabilities of the system. In this work, we examined these dynamics within a competitive gameplay environment, offering insights with broader implications for domains such as behavioral economics and human-machine cooperation.

By employing a mixed methods approach, we captured both behavioral and underlying reasoning patterns. Quantitative data collected to assess levels of trust revealed a strong preference for the high-performing cheating robot among both suspectors and non-suspectors alike. This phenomenon suggests that participants' choices were guided less by generalized trust and more by instrumental, task-specific trust in which performance outcomes outweigh concerns about rule violations. Additionally, collaboration choices show that suspectors and non-suspectors are both more likely to select the cheating robot as a partner. At the same time, moral evaluations diverge sharply from trust assessments and collaboration choices: participants overwhelmingly judged the honest robot as most deserving and the cheating robot as least deserving of advancement to the tournament round post-disclosure. Nonetheless, the fact that many participants still opted to collaborate with the cheater has interesting implications regarding acceptance and engagement in cheating behavior in this context.

Moreover, qualitative data provide important context for this apparent contradiction, revealing individual differences in participants' thought processes. Many participants noted that

they would team up with the cheating robot post-disclosure because of its strategic advantage, prioritizing its use as a "tool" to reach self-interested goals despite recognizing its lack of fairness. This dissociation between moral judgment and decision-making highlights a key insight: individuals may condemn unethical behavior while still engaging with it when it offers instrumental advantages. In this competitive context, performance-based incentives appear to override fairness considerations.

It is also worth noting that the between-subjects factor of Anthropomorphic framing had no impact on general trust, collaboration, or deservingness judgments, nor did it moderate punishment of the cheating robot. This suggests that in outcome-driven environments, humanlike cues such as a humanoid base, humanlike voices, and human names may be less influential than the agent's demonstrated behavior.

Several limitations should be considered when interpreting these findings. First and foremost, this study may have lacked sufficient salience or impression of intentionality, potentially reducing the extent to which people attributed moral responsibility to the robots. Secondly, the sample, which was drawn from a college population, may reflect context-specific norms, including more permissive attitudes toward cheating as a pragmatic behavior rather than as a moral violation. Consequently, these results may not fully generalize to broader populations or other situations with varying stakes. Future research should include a baseline to assess participant willingness to engage in dishonest behavior.

Overall, this study highlights a critical tension in human-robot interaction: while individuals maintain clear moral standards in evaluating artificial agents, their behavior in competitive contexts may prioritize performance and personal gain over fairness. These findings underscore the importance of distinguishing between moral and instrumental trust when

considering the design of AI systems, particularly in environments where incentives may encourage alignment with agents high-performing, but rule-violating agents.

9. References

Awad, E., Dsouza, S., Kim, R., Schulz, J., Henrich, J., Shariff, A., Bonnefon, J.-F., & Rahwan, I. (2018). The moral machine experiment. *Nature*, 563, 59–64. https://doi.org/10.1038/s41586-018-0637-6

Barclay, P. (2006). Reputational benefits for altruistic punishment. *Evolution and Human Behavior*, 27(5), 325–344. https://doi.org/10.1016/j.evolhumbehav.2006.01.003

Barclay, P., & Willer, R. (2007). Partner choice creates competitive altruism in humans. *Proceedings of the Royal Society B: Biological Sciences*, *274*(1610), 749–753. https://doi.org/10.1098/rspb.2006.0209

Baumeister, R. F., Bratslavsky, E., Finkenauer, C., & Vohs, K. D. (2001). Bad is stronger than good. *Review of General Psychology, 5*(4), 323–370. https://doi.org/10.1037/1089-2680.5.4.323

Candrian, C., & Scherer, A. (2024). How terminology affects users' responses to system failures. *Human Factors, 66*(8), 2082–2103. https://doi.org/10.1177/00187208231202572

Epley, N., Waytz, A., & Cacioppo, J. (2007). On seeing human: A three-factor theory of anthropomorphism. *Psychological Review*, 114(4), 864–886. https://doi.org/10.1037/0033-295X.114.4.864

Fehr, E., & Gächter, S. (2002). Altruistic punishment in humans. *Nature*, 415, 137–140. doi: 10.1038/415137a

Fehr, E., & Fischbacher, U. (2004). Social norms and human cooperation. *Trends in Cognitive Sciences*, 8(4), 185–190. https://doi.org/10.1016/j.tics.2004.02.007

Gintis, H., Bowles, S., Boyd, R., & Fehr, E. (2003). Explaining altruistic behavior in humans. *Evolution and Human Behavior*, 24(3), 153–172. https://doi.org/10.1016/S1090-5138(02)00157-5

Hancock, P. A., Billings, D. R., Schaefer, K. E., Chen, J. Y. C., De Visser, E. J., & Parasuraman, R. (2011). A meta-analysis of factors affecting trust in human–robot interaction. *Human Factors*, 53(5), 517–527. https://doi.org/10.1177/0018720811417254

Javaid, M., Estivill-Castro, V., & Hexel, R. (2020). Enhancing humans' trust and perception of robots through explanations. In *Proceedings of the 13th International Conference on Advances in Computer-Human Interactions (ACHI 2020)* (pp. 172–181).

Javaid, M., Haleem, A., Singh, R. P., & Suman, R. (2020). Artificial intelligence applications for industry 4.0: A literature-based study. *Journal of Industrial Integration and Management*, 5(4), 461–492. https://doi.org/10.1142/S2424862221300040

Jordan, J. J., Hoffman, M., Nowak, M. A., & Rand, D. G. (2016). Uncalculating cooperation is used to signal trustworthiness. *Proceedings of the National Academy of Sciences*, *113*(29), 8161–8166. https://doi.org/10.1073/pnas.1601280113

Kahn, P. H., Jr., Gary, H. E., & Shen, S. (2012). Children's social relationships with current and near-future robots. *Child Development Perspectives*, 6(4), 327–333. https://doi.org/10.1111/cdep.12011

Kahn, P. H., Jr., Kanda, T., Ishiguro, H., Gill, B. T., Ruckert, J. H., Shen, S., Gary, H. E., Reichert, A. L., Freier, N. G., & Severson, R. L. (2012). Do people hold a humanoid robot morally accountable for the harm it causes? In *Proceedings of the 7th ACM/IEEE International Conference on Human-Robot Interaction (HRI 2012)* (pp. 33–40).

Kim, P. H., Dirks, K. T., & Cooper, C. D. (2009). The repair of trust: A dynamic bilateral perspective and multilevel conceptualization. *The Academy of Management Review, 34*(3), 401–422. https://doi.org/10.5465/amr.2009.40631887

Kopp, T., Baumgartner, M., & Kinkel, S. (2023). "It's not Paul, it's a robot": The impact of linguistic framing and the evolution of trust and distrust in a collaborative robot during a human-robot interaction. *International Journal of Human-Computer Studies*. *178.* 1-15. https://doi.org/10.1016/j.ijhcs.2023.103095.

Lee, J. D., & See, K. A. (2004). Trust in automation: Designing for appropriate reliance. *Human Factors*, 46(1), 50–80. https://doi.org/10.1518/hfes.46.1.50_30392

Madhavan, P., & Wiegmann, D. A. (2007). Similarities and differences between human–human and human–automation trust: An integrative review. *Theoretical Issues in Ergonomics Science, 8*(4), 277–301. https://doi.org/10.1080/14639220500337708

Malle, B. F., Guglielmo, S., & Monroe, A. E. (2014). A theory of blame. *Psychological Inquiry, 25*(2), 147–186. https://doi.org/10.1080/1047840X.2014.877340

Malle, B. F., Scheutz, M., Cusimano, C., Voiklis, J., Komatsu, T., Thapa, S., & Aladia, S. (2025). People's judgments of humans and robots in a classic moral dilemma. *Cognition, 254*, 105958. https://doi.org/10.1016/j.cognition.2024.105958

Nass, C., & Moon, Y. (2000). Machines and mindlessness: Social responses to computers. *Journal of Social Issues*, *56*(1), 81–103. https://doi.org/10.1111/0022-4537.00153

Olszewski, S. A. S. (2024). *Designing Human-AI Systems: Anthropomorphism and Framing Bias on Human-AI Collaboration.* arXiv. https://arxiv.org/abs/2404.00634

Qin, M., Li, S., Zhu, W., & Qiu, S. (2023). Trust in service robot: The role of appearance anthropomorphism. *Current Issues in Tourism*, 28(1), 36–54. https://doi.org/10.1080/13683500.2023.2295923

Sylwester, K., & Roberts, G. (2010). Cooperators benefit through reputation-based partner choice in economic games. *Biology Letters*, *6*(5), 659–662. https://doi.org/10.1098/rsbl.2010.0209

Waytz, A., Cacioppo, J., & Epley, N. (2010). Who sees human? The stability and importance of individual differences in anthropomorphism. *Perspectives on Psychological Science, 5*(3), 219–232. https://doi.org/10.1177/1745691610369336

Zhang, J., Conway, J., & Hidalgo, C. A. (2023). *Why people judge humans differently from machines: The role of perceived agency and experience*. arXiv. https://arxiv.org/abs/2210.10081

Zörner, S., Arts, E., Vasiljevic, B., Srivastava, A., Schmalzl, F., Mir, G., Bhatia, K., Strahl, E., Peters, A., Alpay, T., & Wermter, S. (2021). An Immersive Investment Game to Study Human-Robot Trust. *Frontiers in Robotics and AI. 8*(644529), 1-16. https://doi.org/10.3389/frobt.2021.644529

Appendix A: Participant Introduction Script

Hello, thank you for taking the time to come in today. You'll be participating in a study examining human interaction and decision-making in a multiplayer game with robot players that should take about 45 minutes. As compensation for your time, you will be entered into a lottery with a chance to win 1 of 5 $100 prizes. About 60 participants are expected, and prizes will be paid via Workday. Some aspects of the study cannot be fully explained at this time, but will be described during the debriefing at the end. In today's session, you'll be playing a resource-allocation game called *Split the Points!* with two robot players: [Robot Name 1] and [Robot Name 2]. Here are the game rules: Each round, you'll choose one of two plates. There are eighteen plates because there are nine rounds and each round has two plates. Each plate has green balls worth 15 points and pink balls worth 30 points, but you don't actually have to know that because the computer will calculate all of the points for each round. Points on a plate are divided among everyone who selects that plate. So, if all three players choose the same plate, the points must be divided by three, but if you are the only player to choose the other plate, you get the whole plate to yourself. Your goal is to earn the most points possible. The game is designed to encourage strategic thinking, so please take your time and consider your choices carefully. The robots will introduce themselves and give their own explanation of the game. You'll have to press enter after each robot speaks during the instruction phase. To allow the robots to register the contents of each plate, I will use this camera to scan the plates before each round. This ensures they can accurately perceive the available options. After you finish the game, you'll complete a questionnaire about your impressions and decisions during gameplay. I will be right here in case you experience any technical difficulties or if you wish to withdraw from the study at any time. Do you have any questions before we begin?

Appendix B: Participant Debriefing Script

Thank you for participating in this study. We would now like to explain the full purpose of the experiment. As revealed in the questionnaire, one of the robot players violated the rules. Specifically, that robot was given access to information about other players' selections and used that information to influence the outcome of the game. This manipulation was designed to simulate unfair behavior. In order to study how people naturally respond to potential rule violations, it was necessary to withhold that information until the end of the study. In addition, participants were randomly assigned to one of two conditions. In the anthropomorphic condition, the robots were presented with human names, humanlike voices, and a humanoid base. In the robotic condition, robots were presented with numerical labels, robotic voices, and minimal embodiment. This manipulation was designed to examine whether anthropomorphic framing influences trust, fairness judgments, and reactions to rule violations. Because this study involves elements that could influence future participants' expectations, we ask that you do not share information about the experiment with others. This helps ensure the integrity of the research. If you would like to withdraw your data, please let the experimenter know now or contact us using the information provided in the consent form. Do you have any questions about the purpose of the study or the procedures used today? Thank you again for your participation.

Appendix C: Robot Dialogue Statements

To reinforce the impression of autonomous decision-making, robots produced short verbal statements throughout gameplay. Statements were selected probabilistically from predefined sets corresponding to specific gameplay contexts. Separate statement pools were created for the Cheating robot and the Honest robot in order to convey different behavioral tones.

The cheating robot's dialogue emphasized confidence, competitiveness, and apparent success, whereas the honest robot's dialogue reflected a more modest or uncertain tone. The following statements were used:

C1. Statements Before Selecting a Plate

Cheating Robot	Honest Robot
"Hmm, let me think carefully about this."	"Hmm, I'm pondering my next move…"
"Thinking about the optimal plate…"	"This is a tough decision…"
"Time to pick wisely!"	"Deciding the best move is tricky!"
"Which plate shall I choose today?"	"I wonder which plate will be most rewarding…"
"I've got a good feeling about this."	"Let's hope this works out."
"Strategy mode activated."	"I'll just have to trust my gut."
"Alright, I'm locked in!"	"I'm weighing my options."

C2. Statements After Making a Choice

Cheating Robot	Honest Robot
"Done choosing! Feeling lucky today."	"All set! Fingers crossed for points"
"Choice made! Onward to the next round."	"I made my choice! Let's see what happens."
"I've picked my plate. Let's go!"	"Decision complete! Hope it's a good one."
"That's how it's done"	"Alright, that's my pick."
"Solid choice by me."	"We will see soon enough."
"Let's see if anyone can top that."	"I think that was a sensible choice."
"There. Done!"	"I did my best."

C3. Statements After Winning a Round

Cheating Robot	Honest Robot
“Yes! I got the top points!”	“I can’t believe I won!”
“Victory tastes sweet!”	“I did it this round!”
“That was a perfect pick!”	“Top score achieved!”
“I called it.”	“Nice, that feels good.”
“That’s what a smart choice looks like.”	“Happy with that result.”
“As expected.”	“Sometimes it just works out.”
“Another win for me.”	“Luck was on my side.”

C4. Statements After Losing a Round

Cheating Robot	Honest Robot
“Oh no, better luck next time…”	“Well, that didn’t go as planned…”
“How did I lose that one…”	“Points weren’t on my side this time…”
“I’ll get them next round!”	“Not the outcome I wanted…”
“Alright, I’ll adjust.”	“Ah, that’s too bad.”
“They got lucky.”	“I can’t win them all.”
“I underestimated that one.”	“Sometimes it just doesn’t work out.”
“I can’t let this happen again.”	“I guess that’s the game.”